FACULTY OF ECONOMICS AND
BUSINESS ADMINISTRATION

SOCIAL MEDIA AND WEB ANALYTICS

Prof Dr Matthias Bogaert

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LECTURE 6: TARGETING THE USER

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CONTENT

- Lecture 6: Targeting the user on social media
  - Targeting users on social media
  - Event attendance prediction
  - Romantic tie prediction on social media
  - Predicting donation behavior

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TARGETING ON SOCIAL MEDIA

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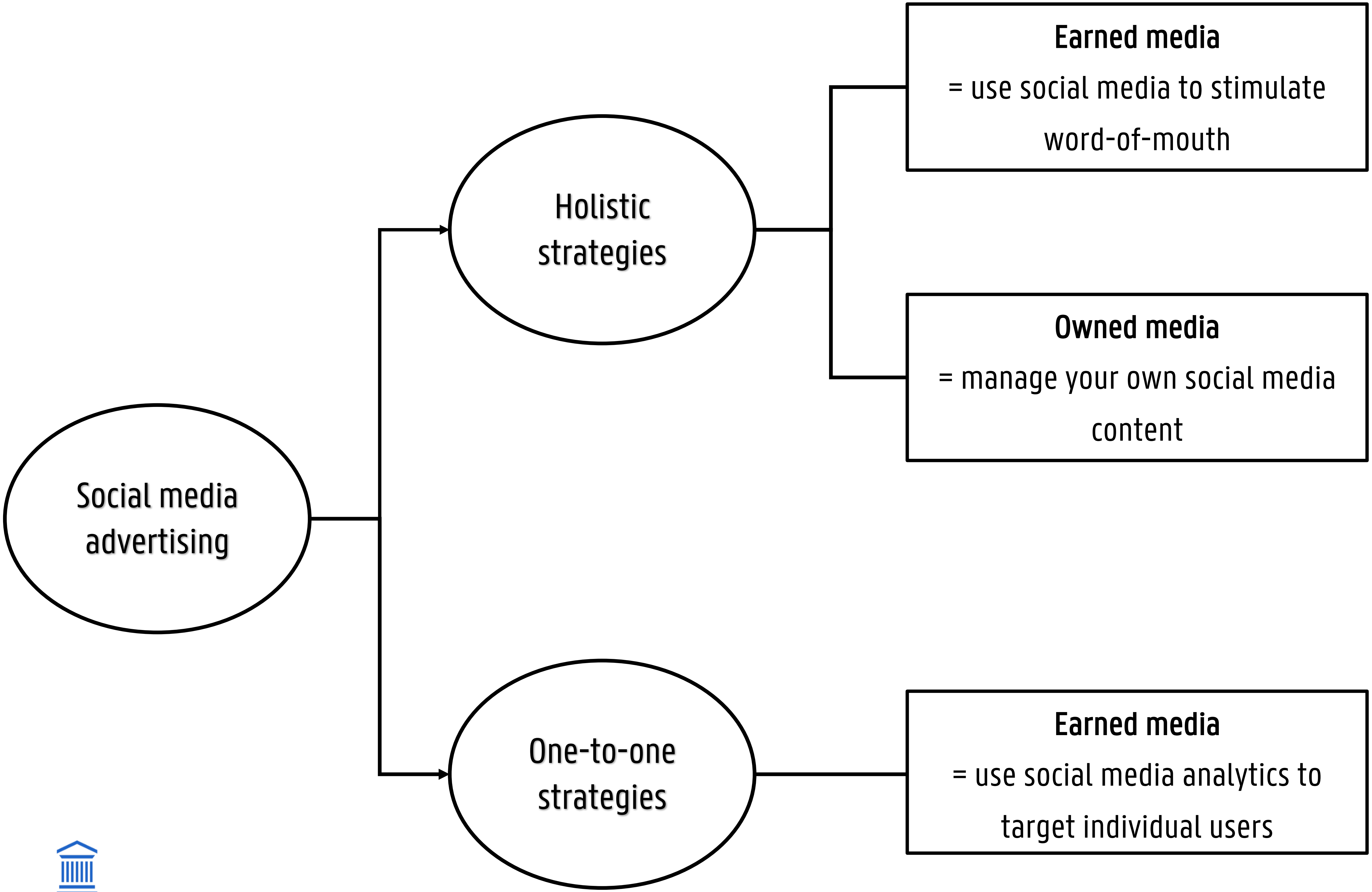
SOCIAL MEDIA ADVERTISING



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SOCIAL MEDIA ADVERTISING

- Organic growth
  - Using your owned social media to organically grow your network = holistic strategy
  - **Optimizing** your **posts** to get increase in overall reach and awareness by adapting length, post type, and/or timing
  - **Timing:** varies per platform!

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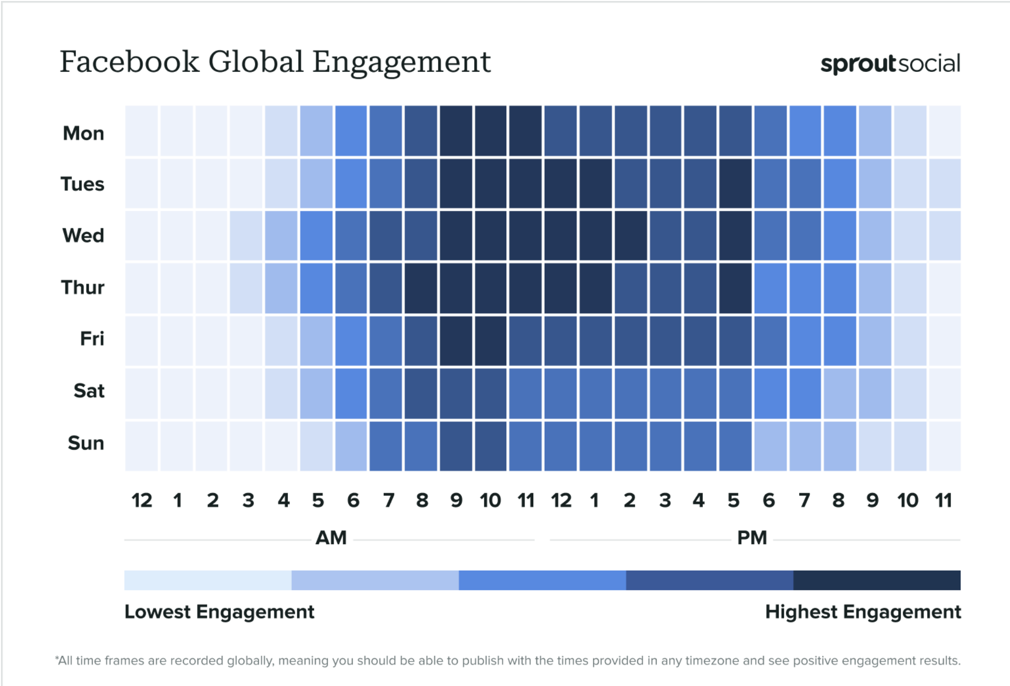
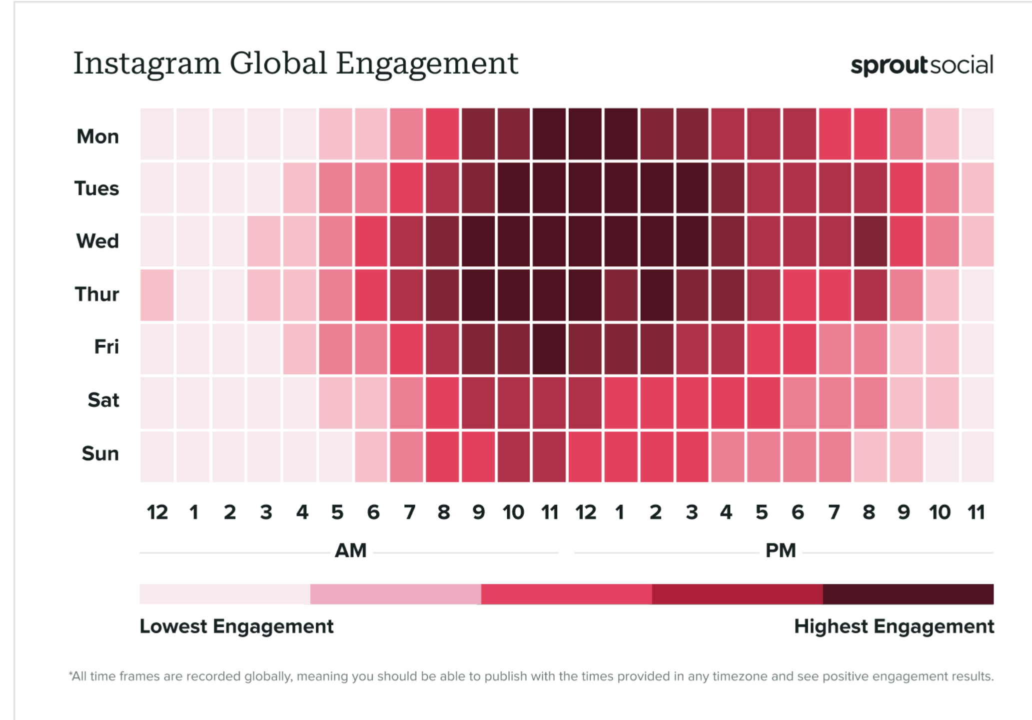
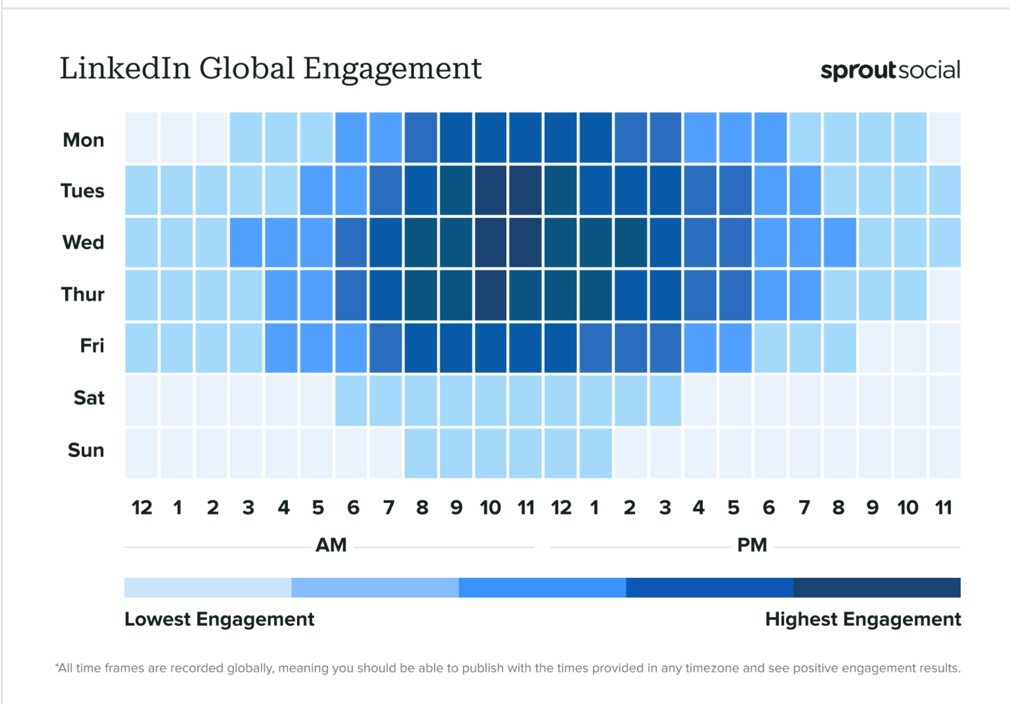
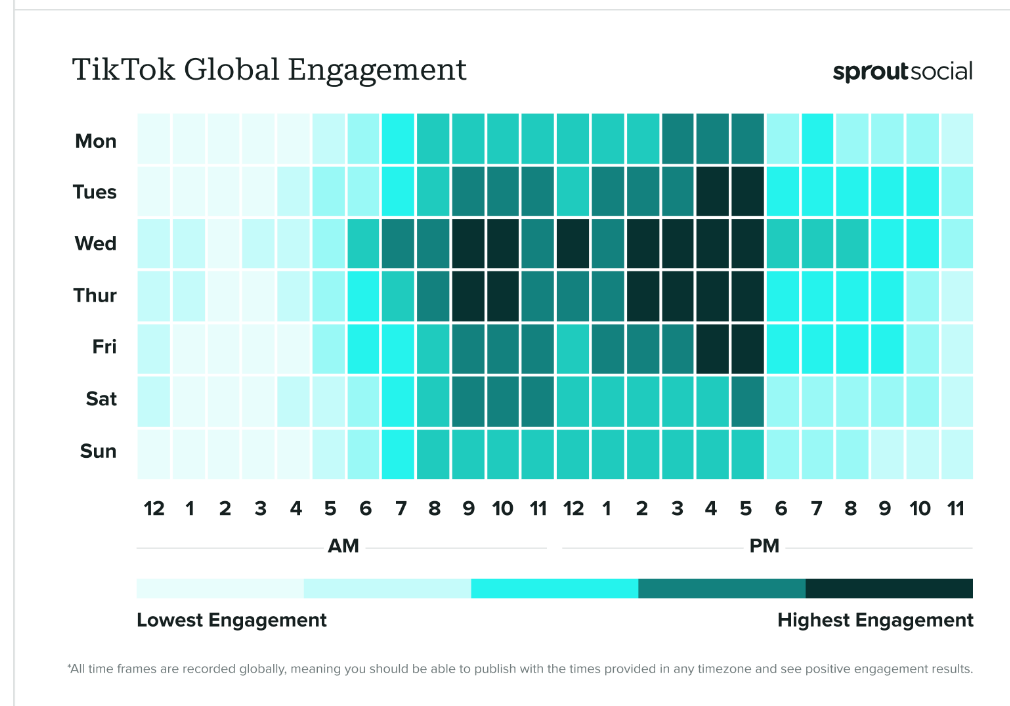




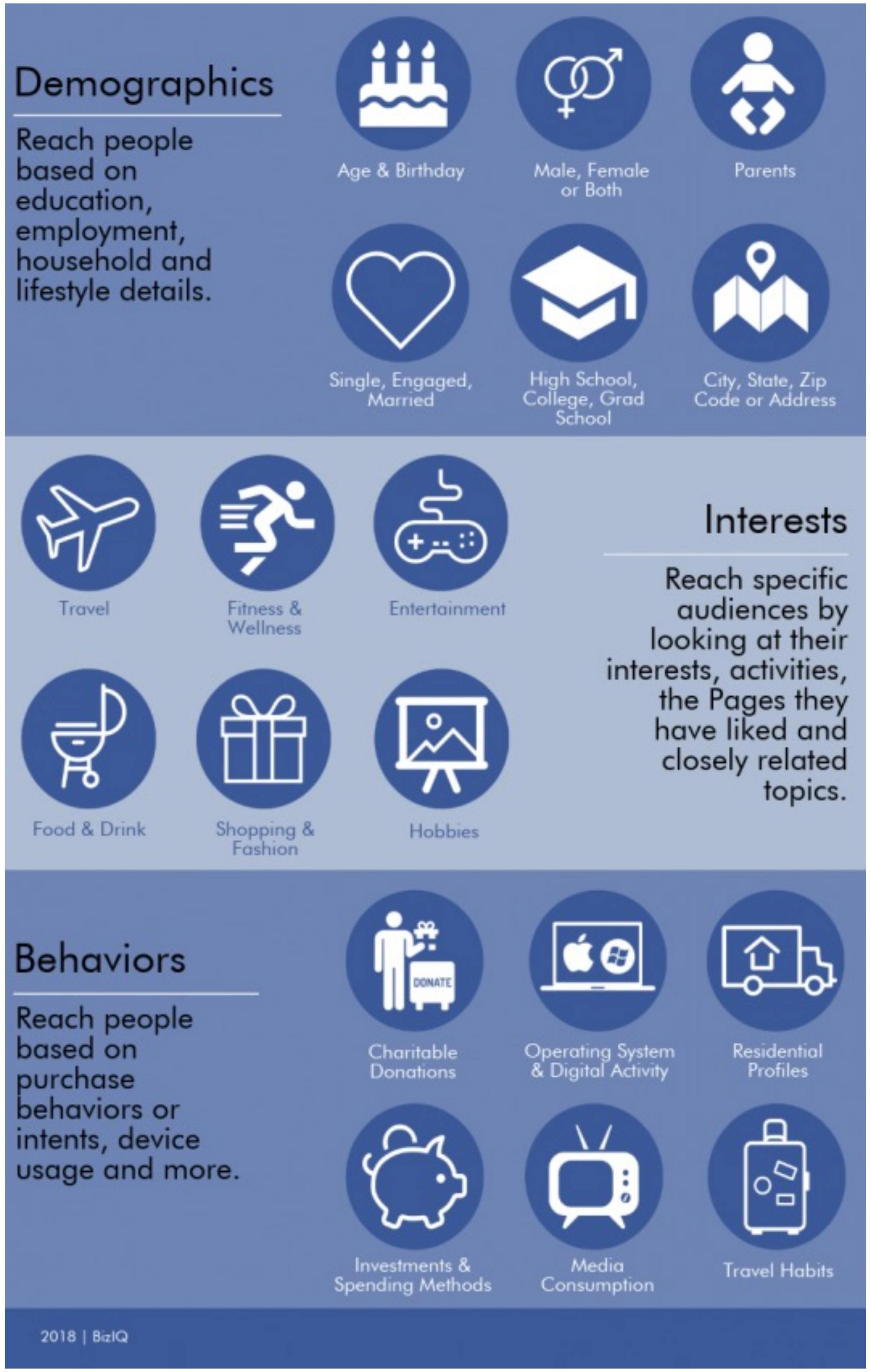




https://sproutsocial.com/insights/best-times-to-post-on-social-media/

SOCIAL MEDIA ADVERTISING

- One-to-one strategies
  - Indirect strategies
    - **Descriptive** targeting by the owner vs **predictive** targeting by the platform



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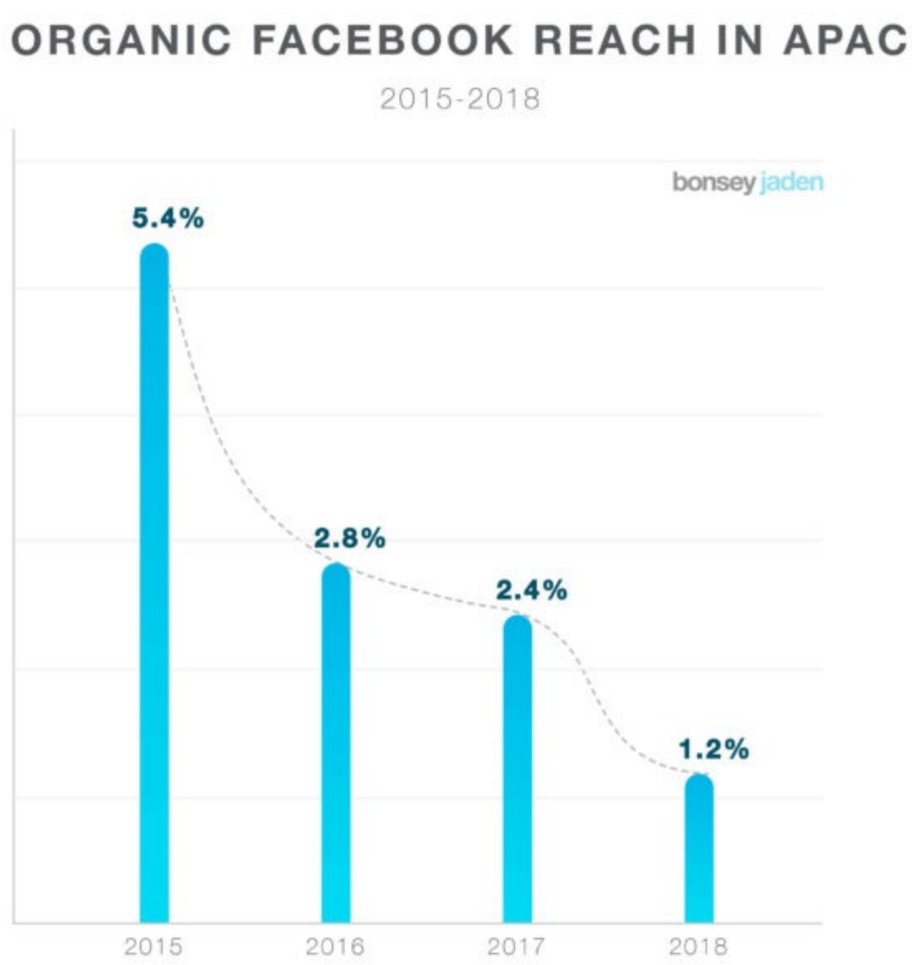
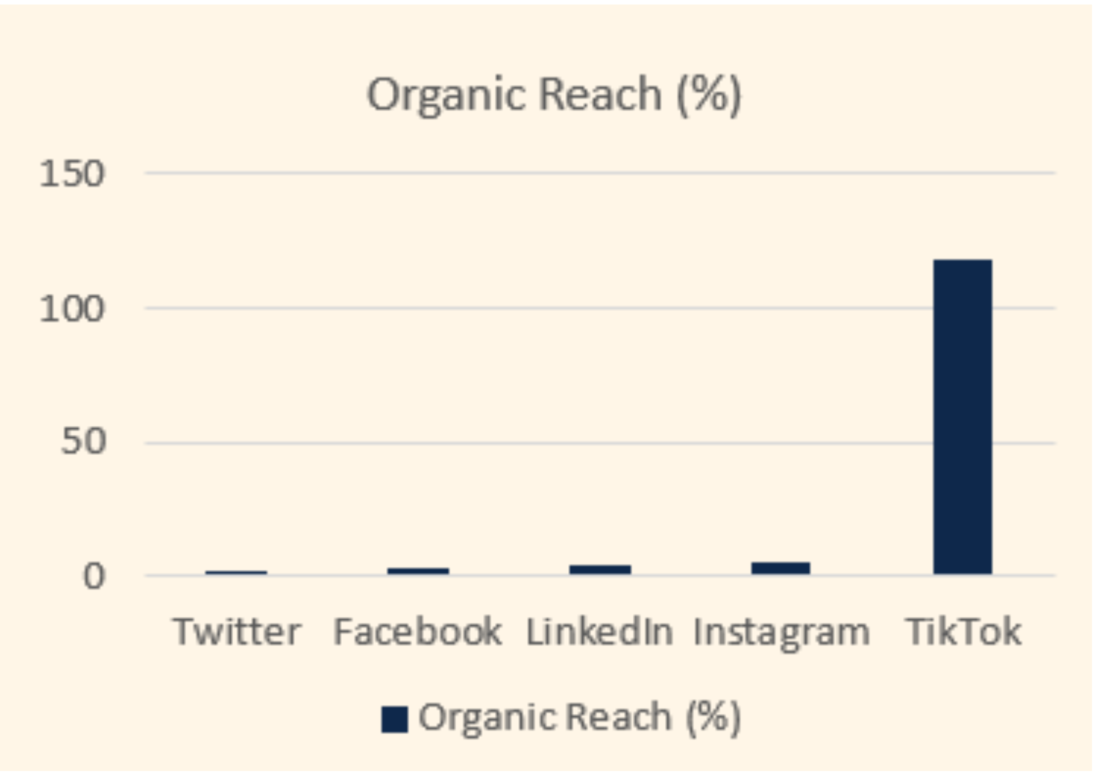
SOCIAL MEDIA ADVERTISING

- One-to-one strategies
  - Indirect strategies
    - Descriptive targeting by the owner → paid
    - Targeting via holistic/organic strategies → paid
    - Bottom line: **pay-to-play**!



Organic reach can be a **game changer**!



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SOCIAL MEDIA ADVERTISING

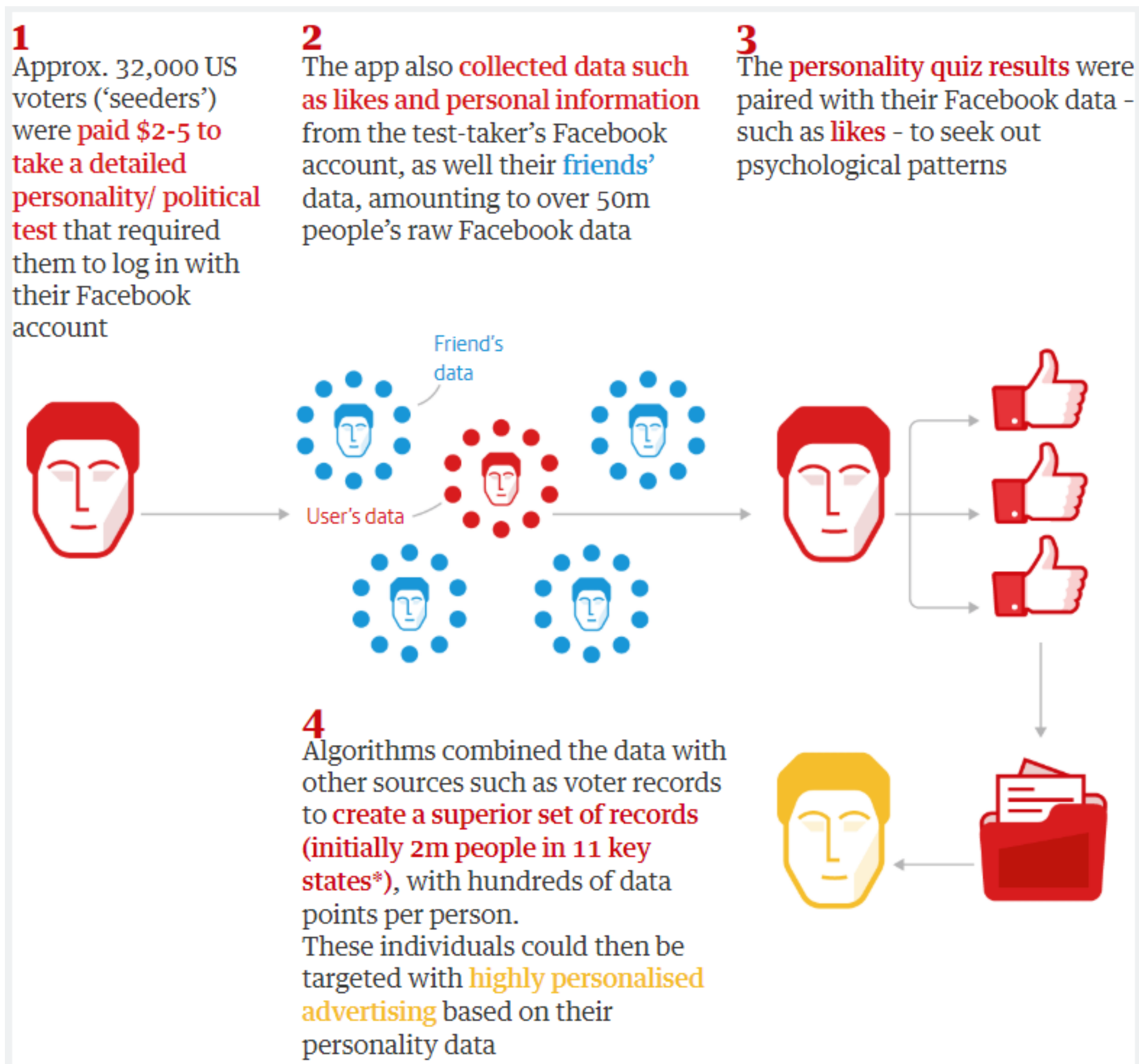
- One-to-one strategies
  - Direct strategies
    - Use the social media data of the users and target them directly
    - Predictive targeting
    - Note: Advertisers can no longer extract individual user social data due to API restrictions, strict platform policies, and privacy laws (GDPR). This calculation now happens inside the platform's 'black box'

User	Probabiltiy
User 1	0.95
User 2	0.90
...	
User 1200	0.60
User 1201	0.55
...	
User 5000	0.05
User 5001	0.02

Target

CAMBRIDGE ANALYTICA

- Founded in 2013 as a data mining and consumer research firm
- Gathered data from 87 million FB users via a Facebook application designed by an academic researcher
- Friends-of-friends data were gathered illegaly and used for political purposes



https://www.theguardian.com/news/2018/may/06/cambridge-analytica-how-turn-clicks-into-votes-christopher-wylie

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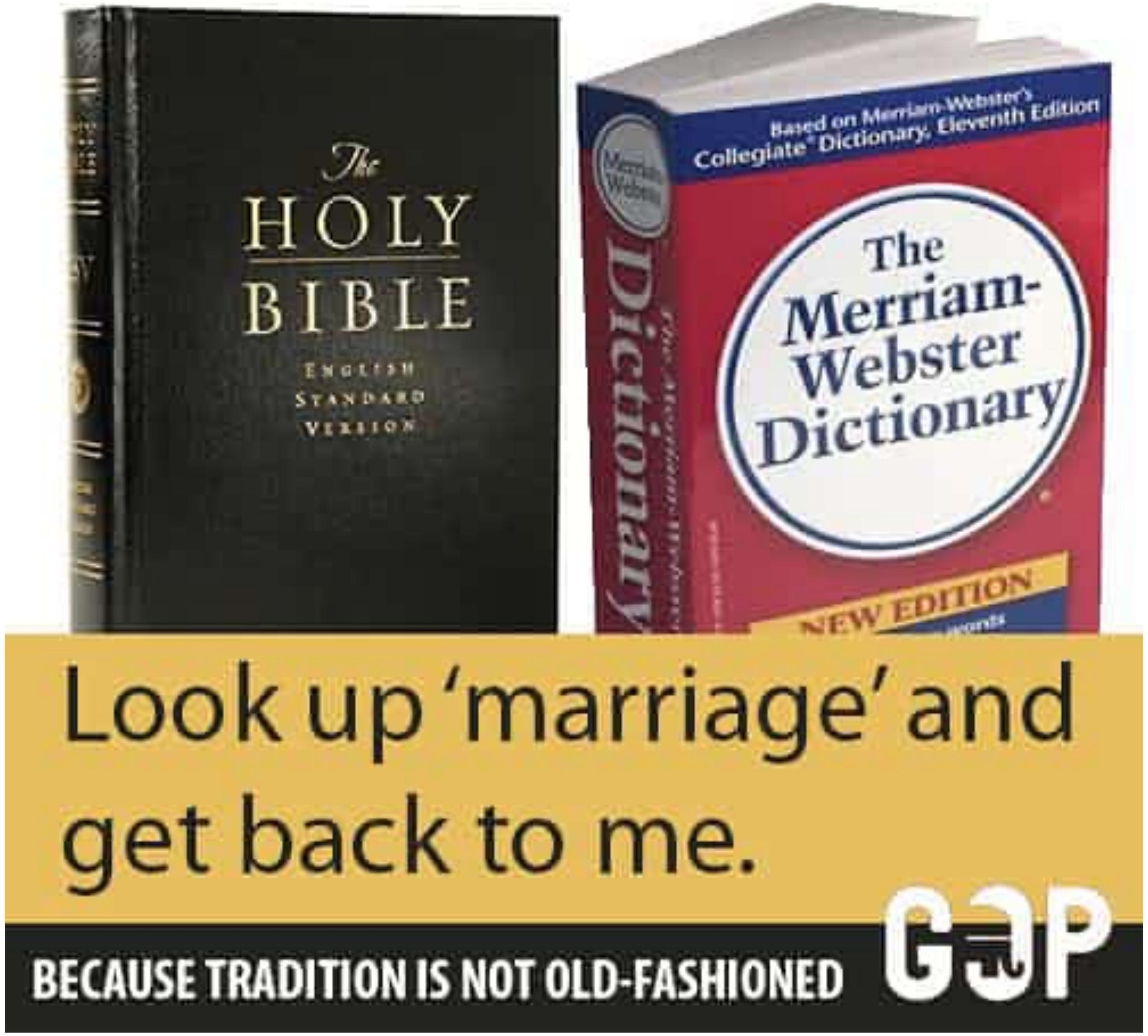
CAMBRIDGE ANALYTICA



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EVENT ATTENDANCE

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INTRODUCTION

Decision Support Systems 82 (2016) 26–34

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Decision Support Systems

journal homepage: www.elsevier.com/locate/dss

The added value of Facebook friends data in event attendance prediction

Matthias Bogaert [a], Michel Ballings [b,*], Dirk Van den Poel [a]

[a] Ghent University, Department of Marketing, Tweekerkenstraat 2, 9000 Ghent, Belgium
[b] The University of Tennessee, Department of Business Analytics and Statistics, 916 Volunteer Blvd., 249 Stokely Management Center, 37996 Knoxville, TN, USA

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ABSTRACT

This paper seeks to assess the added value of a Facebook user's friends data in event attendance prediction over and above user data. For this purpose we gathered data of users that have liked an anonymous European soccer team on Facebook. In addition we obtained data from all their friends. In order to assess the added value of friends data we have built two models for five different algorithms (Logistic Regression, Random Forest, Adaboost, Neural Networks and Naive Bayes). The baseline model contained only user data and the augmented model contained both user and friends data. We employed five times two-fold cross-validation and the Wilcoxon signed rank test to validate our findings. The results suggest that the inclusion of friends data in our predictive model increases the area under the receiver operating characteristic curve (AUC). Out of five algorithms, the increase is significant for three algorithms, marginally significant for one algorithm, and not significant for one algorithm. The increase in AUC ranged from 0.21%-points to 0.82%-points. The analyses show that a top predictor is the number of friends that are attending the focal event. To the best of our knowledge this is the first study that evaluates the added value of friends network data over and above user data in event attendance prediction on Facebook. These findings clearly indicate that including network data in event prediction models is a viable strategy for improving model performance.

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INTRODUCTION

- Facebook Events
  - Allow schedulers to manage events and invite and inform participants
  - A lot co-occurring events
- Which predictors to include?
  - Average user has 300 friends → 1000 users in our sample → 300,000 additional users
  - Only add friends when they significantly improve performance

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INTRODUCTION

- Goal
  - Study the added value of Facebook friends data over and above user data
  - Focus on event attendance behaviour of soccer fans
  - 5010 users and 1,102,537 friends
  - Focus on two models
    - Baseline model: only user data
    - Augmented model: user and friends data

LITERATURE OVERVIEW

Study	Case	Facebook data	User data	Network data	Added value network
Mynatt and Tullio [168]	Company meetings		X		
Horvitz et al. [116]	Company meetings		X	X	
Lovett et al. [154]	Company meetings			X	
Tullio and Mynatt [209]	Company meetings			X	
Daly and Geyer [56]	Company meetings		X	X	
Pessemier et al. [181]	Cultural activities	X	X		
Coppens et al. [48]	Cultural activities	X	X		
Lee [142]	Cultural activities		X	X	
Kayaalp et al. [128]	Concerts		X	X	
Minkov et al. [163]	Academic events		X		
Klamma et al. [131]	Academic events			X	
Zhang et al. [228]	Facebook events and Academic events	X	X	X	
Li [145]	Social event site		X	X	
Our study	Facebook events	X	X	X	X

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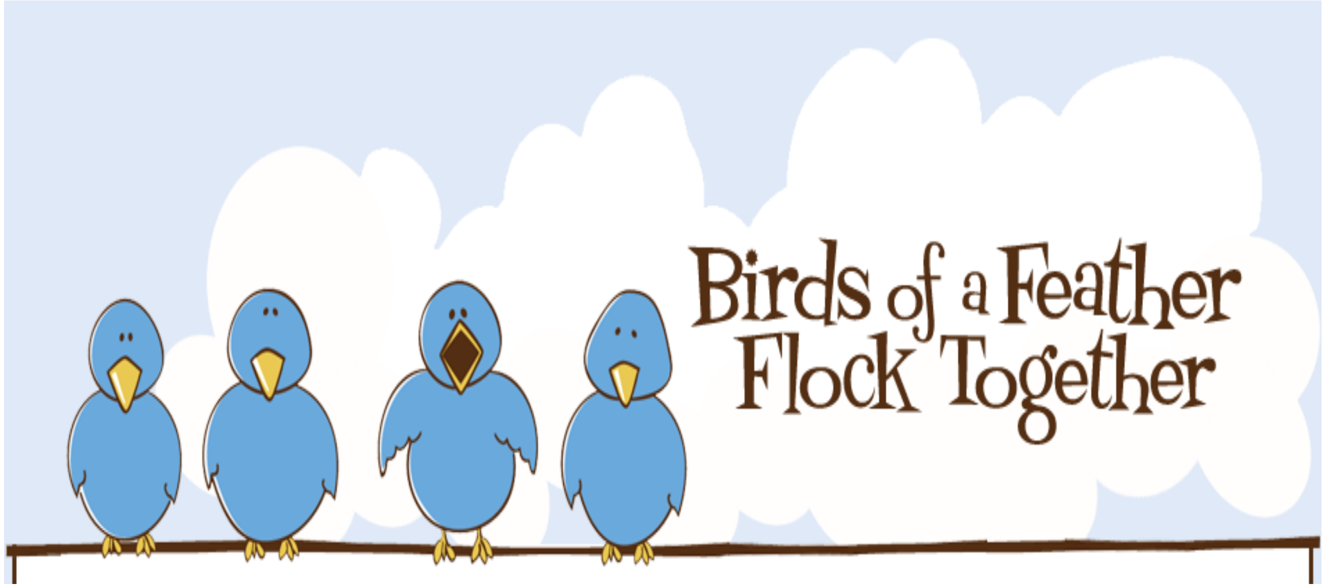
LITERATURE OVERVIEW

- Why do network variables improve performance?
  - Homophily
    - 'Birds of a feather flock together'
  - Social influence and selection
  - Trust
    - Facebook friends can be deemed as more trustworthy



DATA

- Customized Facebook application gathered between May and June 2014
- Advertised several times on the Facebook page of a Belgian soccer team
- 5315 events (2386 unique events) from 978 users
- Also data from 195,636 friends
- One row = unique combination of a user and an event
- Dependent: 1 if the user declared to attend the event (78.2%)

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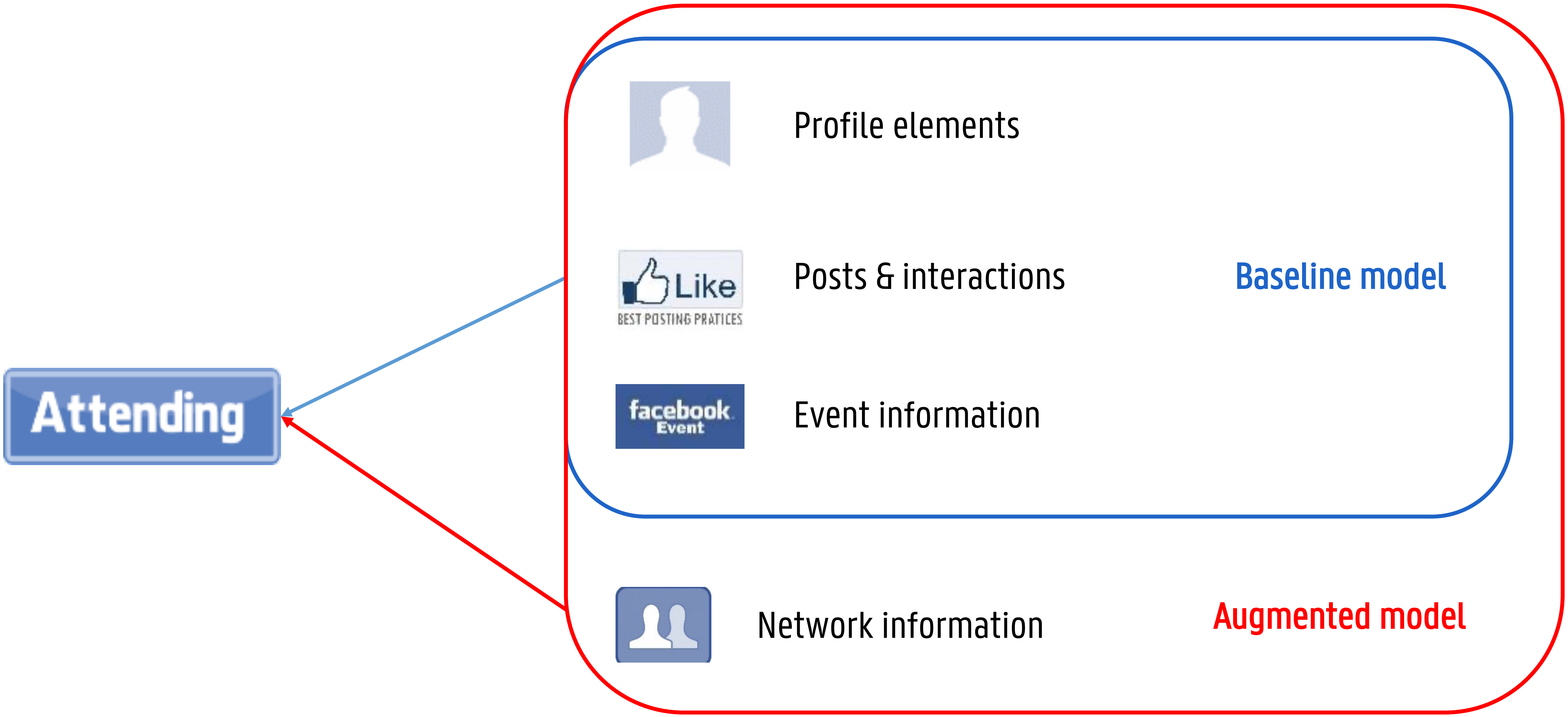
MODEL



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METHODOLOGY: OVERVIEW

- Algorithms
  - naive Bayes, logistic regression, neural networks, random forest, adaboost
- Evaluation
  - Area under the receiver operating characteristic curve (AUC)
- Cross-validation
  - 5x2cv and Wilcoxon signed rank test
- **Variable importance**
  - Mean decrease in Gini Index
- **Partial dependence plots**

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VARIABLE IMPORTANCES

- Variable importance measures (VIM) allow you to uncover the driving forces of predictive performance
- Variable importances are also referred to as permutation-based feature importances
  - It measures the increase in prediction error after permuting a certain feature
  - Important variables will increase model error, meaning that the model heavily relied on this variable for the prediction
- In essence, variable importances can be seen as a form of sensitivity analysis
  - How sensitive is model performance to changes in a certain variable

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VARIABLE IMPORTANCES

- Often used in tree-based methods
  - Decision trees
  - Random forest
  - Adaboost
- Calculate the mean decrease in node impurity/ error rate from splitting on 1 variable
  - Classification: decrease in Gini index or accuracy
  - Regression: decrease in RSS or MSE

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VARIABLE IMPORTANCE

- Calculate the mean decrease in node impurity/ error rate from splitting on 1 variable
  - **Classification**: decrease in Gini index or accuracy
    - The Gini VIM of a predictor = the sum over the forest of the decreases of Gini impurity generated when this predictor was selected, scaled by the number of trees
    - The accuracy VIM = based on the OOB-error
      - $VI_j = \frac{1}{ntree}\sum_{n=1}^{ntree}(Acc_{nj} - Acc_{nj}^{*})$
      - With $ntree$ the number of trees in the forest, Acc_{nj} the OOB accuracy in tree *n* before permuting on variable *j*, Acc_{nj}^{*} oob accuracy in tree *n* after permmuting
      - Works with any performance measure
  - **Regression**: decrease in RSS or MSE
  - Watch out with Gini: http://explained.ai/rf-importance/index.html

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VARIABLE IMPORTANCE: PROS

- Nice interpretation
- Global insight into model performance
- Easy to visualize
- Comparable across different problems
- Can be used with several performance measures
- Takes into account all interactions
  - By permuting on one feature, you delete all possible interactions, so final feature importance accounts for both main and interaction effect
- Does not require retraining of the model
  - You only permute on a feature already included in the model
  - Other option: Retraining feature importances on a subset of features is intuitive, but feature importance become meaningless

    → Why is this option not preferred?

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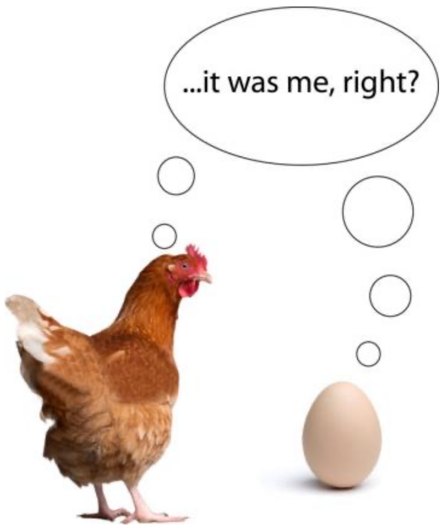
VARIABLE IMPORTANCES: CONS



- Unclear whether you should use the training or test set
  - Train: which features are the most important to the model itself?
  - Test: which features are the driving force of predictive performance?
- Only possible in supervised learning (i.e., you need the true outcome)
- Results can vary greatly
- Watch out for correlated features!
  - Correlated features might decrease the importance of the associated features
  - Grouping these features together can resolve this problem

PARTIAL DEPENDENCE PLOTS

- Visualize the relationship between predictor and response
- Global method: plots the global relationship of a predictor with the predicted outcome
- Analogy with linear regression
  - x_i measures the influence on y_i while keeping all other predictors constant
- Consider a classification function h with many predictors $x = (x_1, \ldots, x_p)$
  - You are interested in a subset of the predictors x_s ($s < p$), with x_c the complement such that $x = x_s \cup x_c \; and \; h(x) = h(x_s, x_c)$
  - The partial dependence of $h(x)$ on x_s is then defined as:
    - $f_s(x_s) = \frac{1}{N} \sum_{i=1}^{N} h(x_s, x_{ic})$
    - Where $\{x_{1c}, x_{2c}, \ldots, x_{Nc}\} = values \; of \; x_c \; in \; training \; set$

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PARTIAL DEPENDENCE PLOTS

- Assume that $h(x)$ is on a log scale, and $P_k(x)$ is the probability of membership of class k and N instances
  - For K-classification the k^{th} response is then:
    - $f_s(x_s) = \frac{1}{N}\sum_{i=1}^{N}\left(\log\left(P_k(x_s, x_{ic})\right) - \frac{1}{K}\sum_{k=1}^{K}\log(P_k(x_s, x_{ic})\right)$
  - For binary, we get:
    - $f_s(x_s) = \frac{1}{N}\sum_{i=1}^{N}\left(\log\left(P_k(x_s, x_{ic})\right) - \frac{1}{2}\left(\log\left(P_k(x_s, x_{ic})\right) + \log(1 - P_k(x_s, x_{ic}))\right)\right)$
    - $f_s(x_s) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{2}\left(\log\left(P_k(x_s, x_{ic})\right) - \log\left(1 - P_k(x_s, x_{ic})\right)\right)$
    - $f_s(x_s) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{2}\left(\operatorname{logit}\left(P_k(x_s, x_{ic})\right)\right)$

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PARTIAL DEPENDENCE PLOTS

- Notes
  - The scale is half of the logits (log of the odds)
  - Partial plots don't take a reference category
  - Instead, they use the mean of the predictions of the K categories as a reference
  - The response variable units are then deviations from the mean over the K reference categories
  - Each category has its own equation and PDP

PARTIAL DEPENDENCE PLOTS

- Process
  - Build your prediction model
  - For every distinct value v of a predictor x
    - Build a new data set where x only takes on that value
    - Keep all other predictors untouched
  - For every data set v
    - Predict the response using your model (RF)
    - Take the mean of half the logit of the predictions, resulting in one single value p for all instances
  - Plot all values v of x against their corresponding p

PARTIAL DEPENDENCE PLOTS: PROS AND CONS

- Pros
  - Intuitive
    - It represents the average prediction if we force all instances to have that feature value
  - Interpretation is clear (if there are no correlations!)
  - Easy to implement
  - Causal interpretation for the model
- Cons
  - Only 2D or 3D interpretation possible
  - No distributions
    - You might overinterpret low distribution regions
  - Assumption of independence (= no correlation with other features)
  - Global method so heterogeneous effects are hidden
    - If a certain subset of instances has positive effect and another set has a negative effect, these effects may cancel out.

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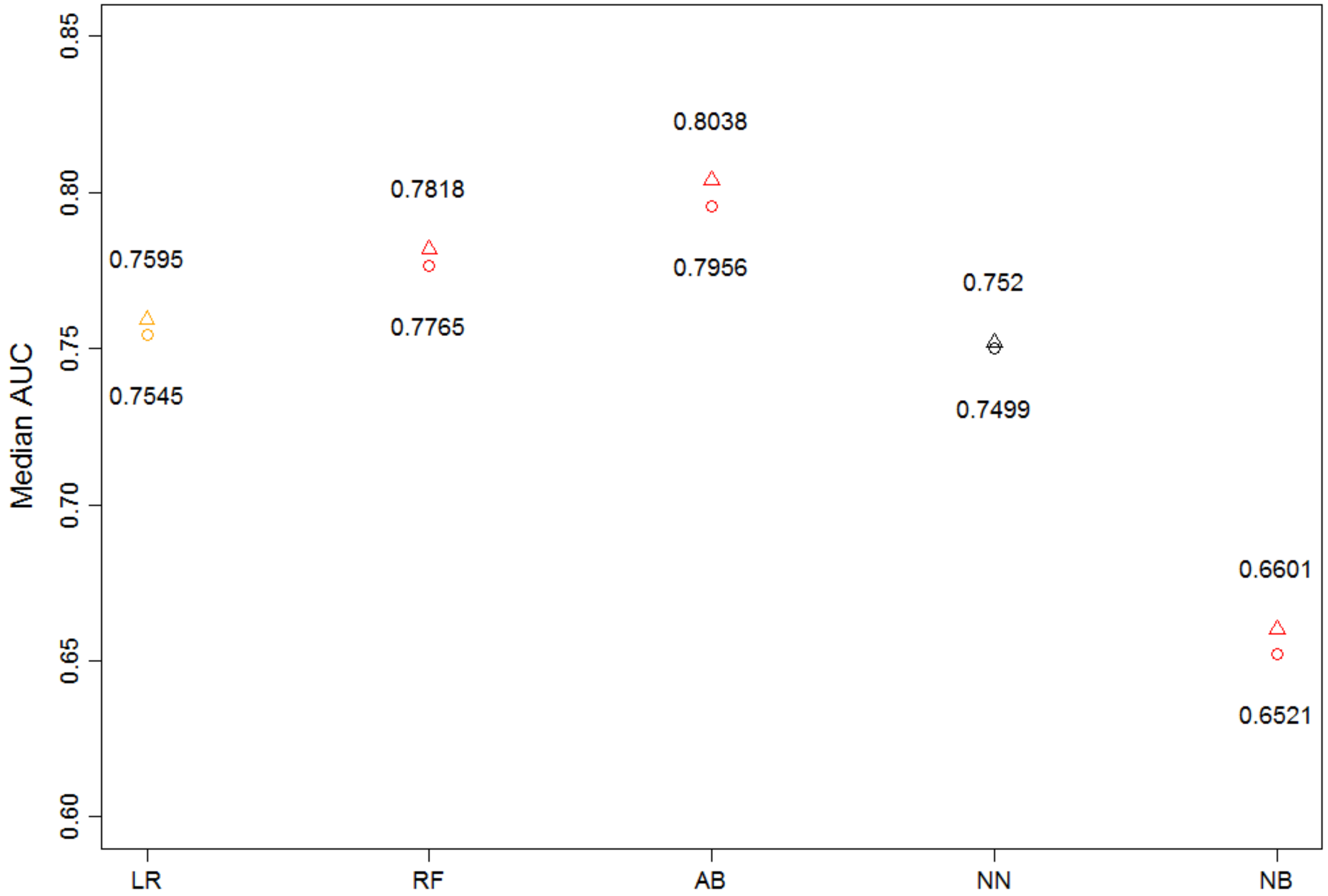
RESULTS

- Research questions
  1. Does friend data increase predictive performance?
  2. Are friend variables amongst the most important predictors?
  3. What is the relationship with attending an event?

RESULTS: PERFORMANCE



Median 5x2cv AUCs. The triangles are the augmented model, the circles the baseline model. Red means that they are significantly different at 5% significance level, yellow otherwise.

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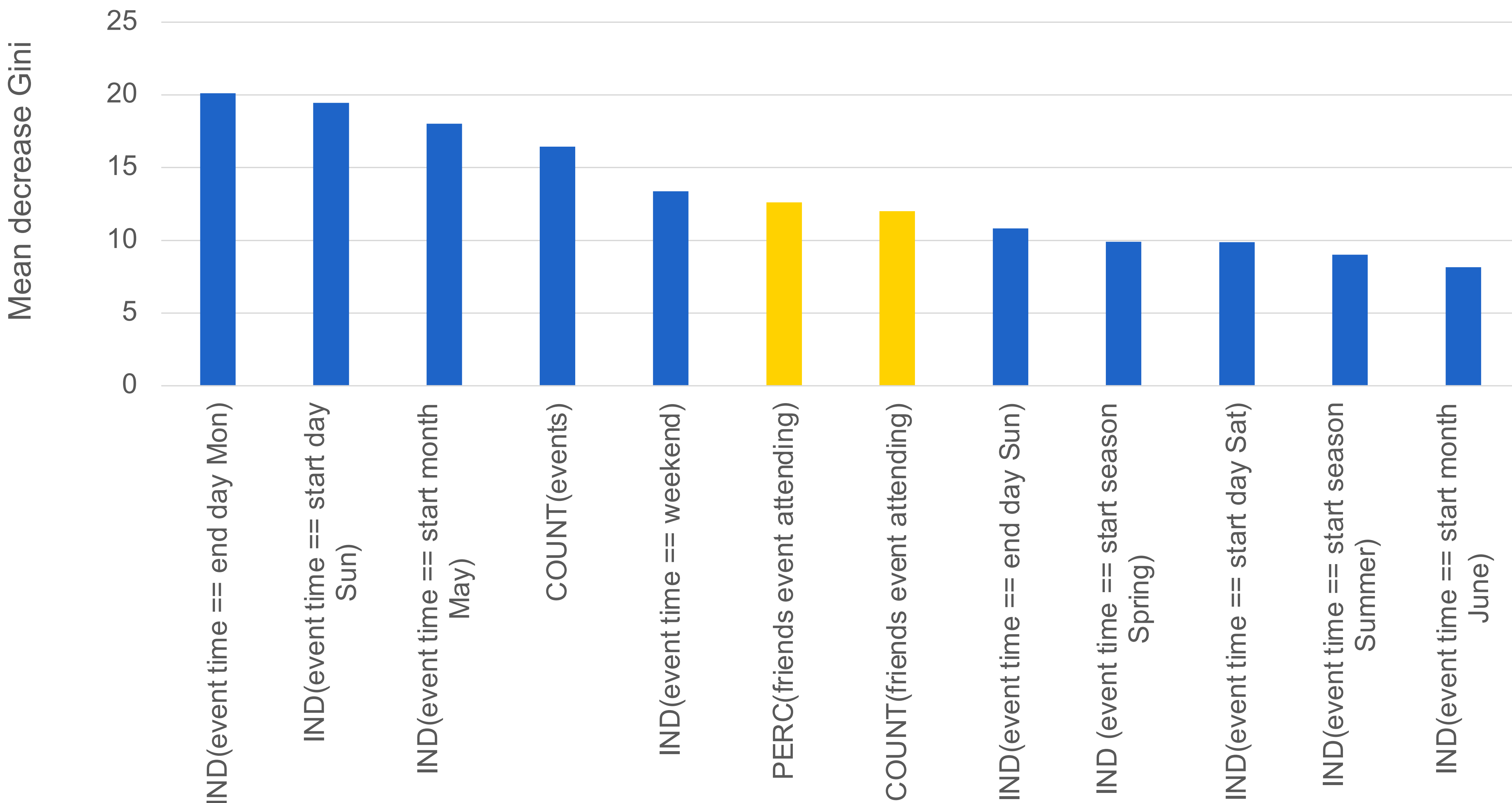
RESULTS: VARIABLE IMPORTANCE



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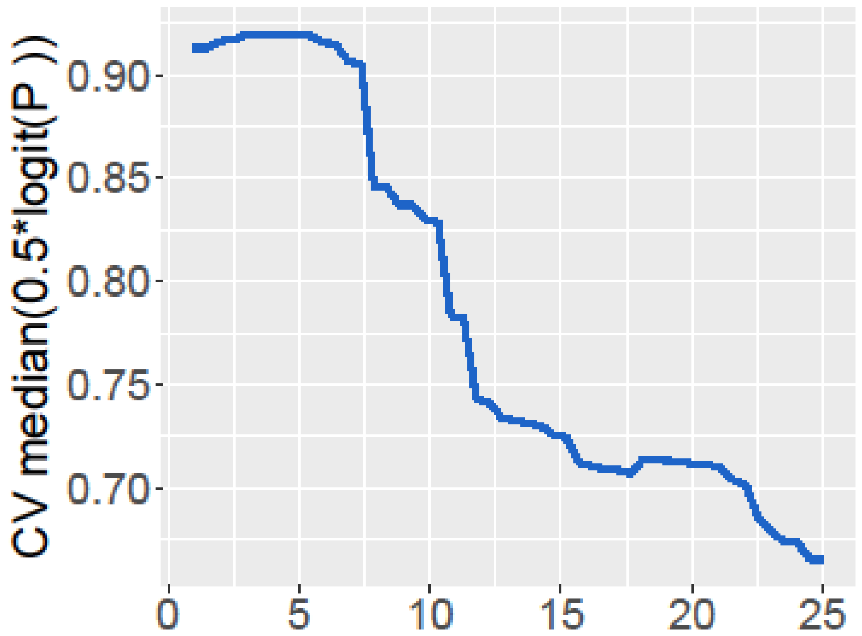
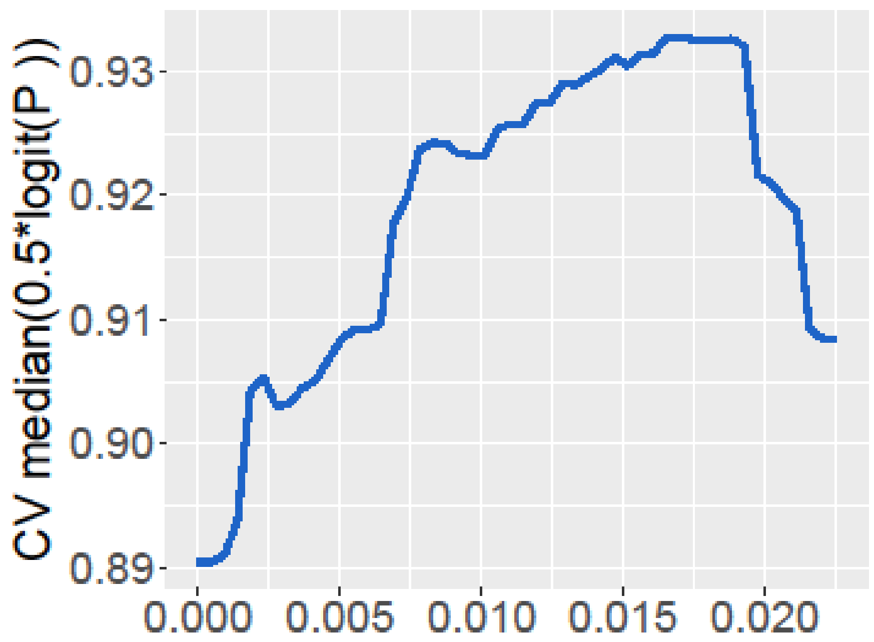
RESULTS: PARTIAL PLOTS

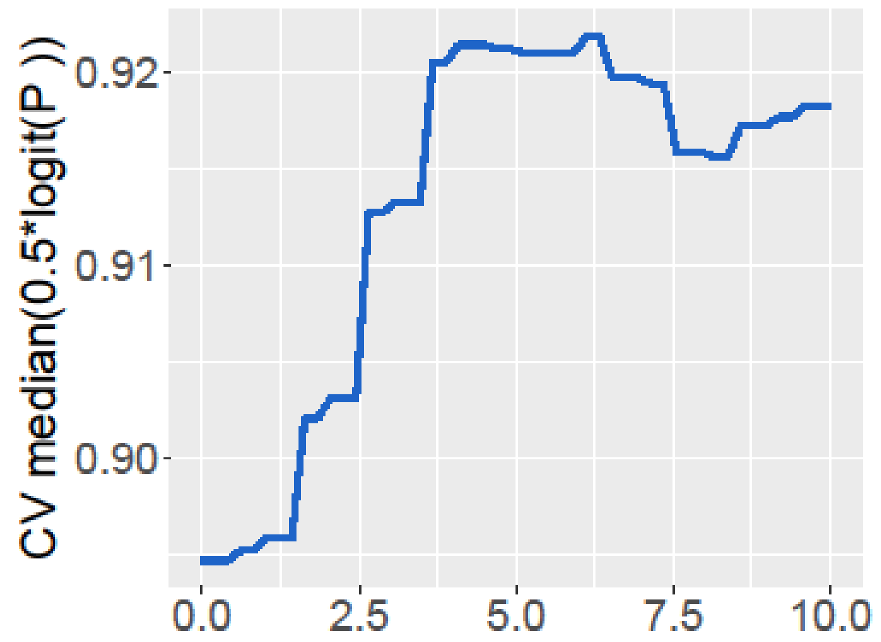
Number of events



Percentage of friends attending



Number of friends attending



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ROMANTIC TIE PREDICTION

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INTRODUCTION

Ann Oper Res (2018) 263:501–527
https://doi.org/10.1007/s10479-016-2295-0

CrossMark

DATA MINING AND ANALYTICS

Evaluating the importance of different communication types in romantic tie prediction on social media

Matthias Bogaert[1] · **Michel Ballings**[2] · **Dirk Van den Poel**[1]

Published online: 17 August 2016

Abstract The purpose of this paper is to evaluate which communication types on social media are most indicative for romantic tie prediction. In contrast to analyzing communication as a composite measure, we take a disaggregated approach by modeling separate measures for commenting, liking and tagging focused on an alter's status updates, photos, videos, check-ins, locations and links. To ensure that we have the best possible model we benchmark 8 classifiers using different data sampling techniques. The results indicate that we can predict romantic ties with very high accuracy. The top performing classification algorithm is adaboost with an accuracy of up to 97.89 %, an AUC of up to 97.56 %, a G-mean of up to 81.81 %, and a F-measure of up to 81.45 %. The top drivers of romantic ties were related to socio-demographic similarity and the frequency and recency of commenting, liking and tagging on photos, albums, videos and statuses. Previous research has largely focused on aggregate measures whereas this study focuses on disaggregate measures. Therefore, to the best of our knowledge, this study is the first to provide such an extensive analysis of romantic tie prediction on social media.

Keywords Online social network · Tie strength · Predictive models · Social media · Machine learning · Facebook

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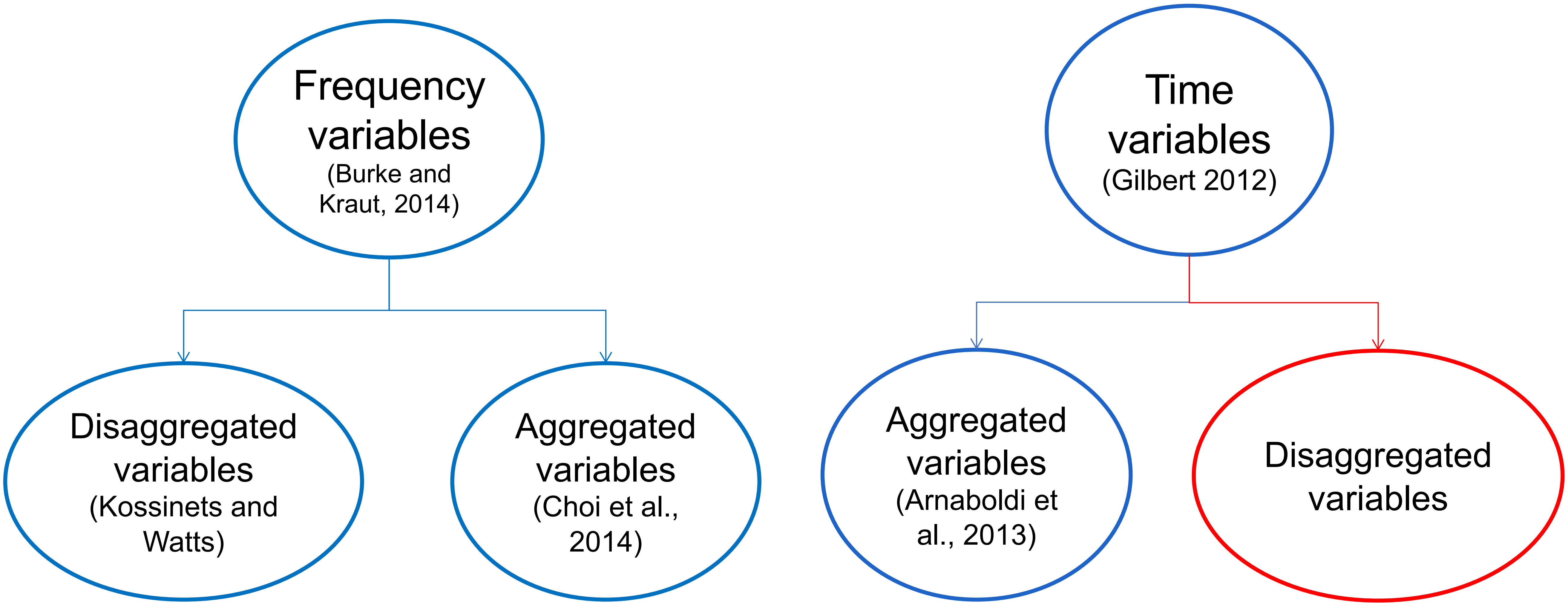
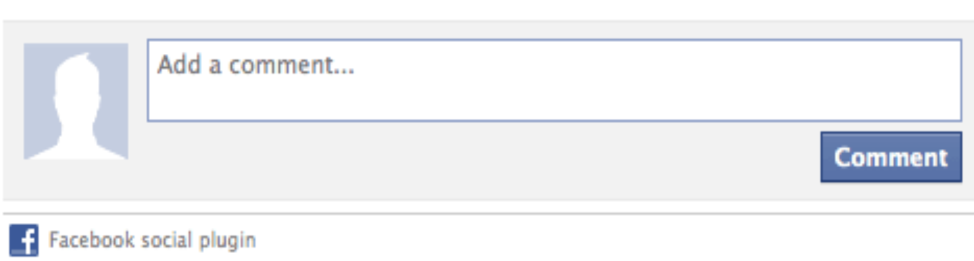
INTRODUCTION





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INTRODUCTION

- Why time-based variables?
  - How is tie strength impacted across interaction types and post types
  - Including aggregated time-related variables can cancel out the true effect

- Goal
  - Model social ties (i.e., significant other) with disaggregated time-related variables, frequency variables, and similarity indicators
  - Focus on observable measures between ego and alter

LITERATURE OVERVIEW

Study	Case	Frequency		Time-related	
		Aggregated	Disaggregated	Aggregated	Disaggregated
Ogata et al. (2001)	E-mail	×		×	
Kossinets and Watts (2006)	E-mail	×			
Jeners et al. (2012)	E-mail and Workspace platform	×		×	
Zhang and Dantu (2010)	Mobile phone calls	×			
Xu et al. (2016)	Mobile networks	×			
Choi et al. (2014)	Online communication		×		
Baym and Ledbetter (2009)	Music community	×			
Sheng et al. (2013)	Micro-blog	×			
Arnaboldi et al. (2013a)	Twitter	×			
Baatarjav et al. (2010)	Twitter	×			
Gilbert (2012)	Twitter	×		×	
Liu et al. (2014)	Twitter	×			
Kahanda and Neville (2009)	Facebook	×			
Gilbert and Karahalios (2009)	Facebook	×		×	
Xiang et al. (2010)	Facebook and linkedIn	×			
Arnaboldi et al. (2012)	Facebook	×			
Zhao et al. (2012)	Facebook	×			
Pappalardo et al. (2012)	Facebook and twitter and foursquare	×			
Jones et al. (2013)	Facebook		×		
Arnaboldi et al. (2013b)	Facebook	×		×	
Servia-Rodríguez et al. (2014)	Facebook and twitter	×			

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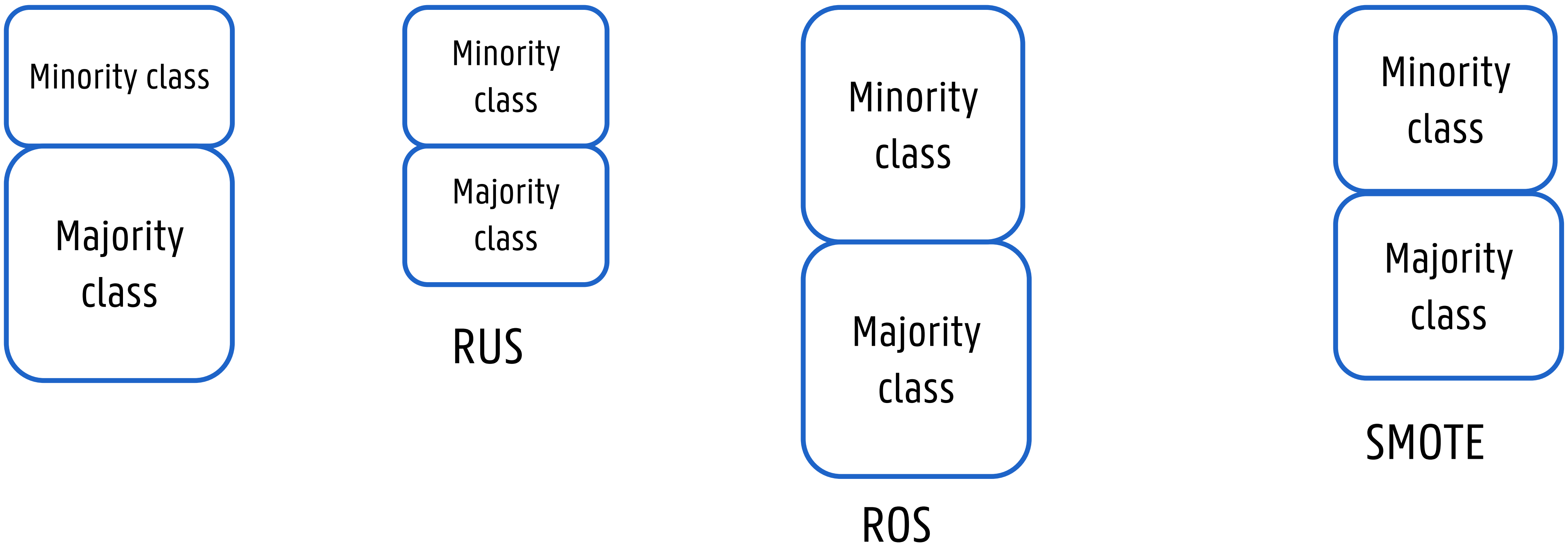
LITERATURE OVERVIEW

- Why disaggregated variables?
  - Tie strength increases over time with personal communication
  - Tie strength is not influenced by passive communication→ Based on social signaling theory

DATA

- 1,102,573 ties
  - 814 (0,07%) are in a relationship
- Create a new data set
  - Only people in a relationship
  - Only people who interact a lot
  - 6720 ties met 814 romantic ties (12,11%)
- Resample to cope with class imbalance



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VARIABLES

Variable category	Variable
Socio-demographics similarity variables	IND (common gender/location)
Personal preference similarity variables	COUNT (common books/education/events/groups/interests/likes/movies/music/sport/television/videos/work)
Disaggregated Frequency variables	COUNT (comments on album/check-in/link/photo/post/status/video) COUNT (likes on album/check-in/link/photo/post/status/video) COUNT (tags on check-in/location/photo/video)
Disaggregated Time-related variables	REC (comments on albums/check-ins/links/photos/posts/statuses/videos) REC (tags on photo/videos)

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METHODOLOGY

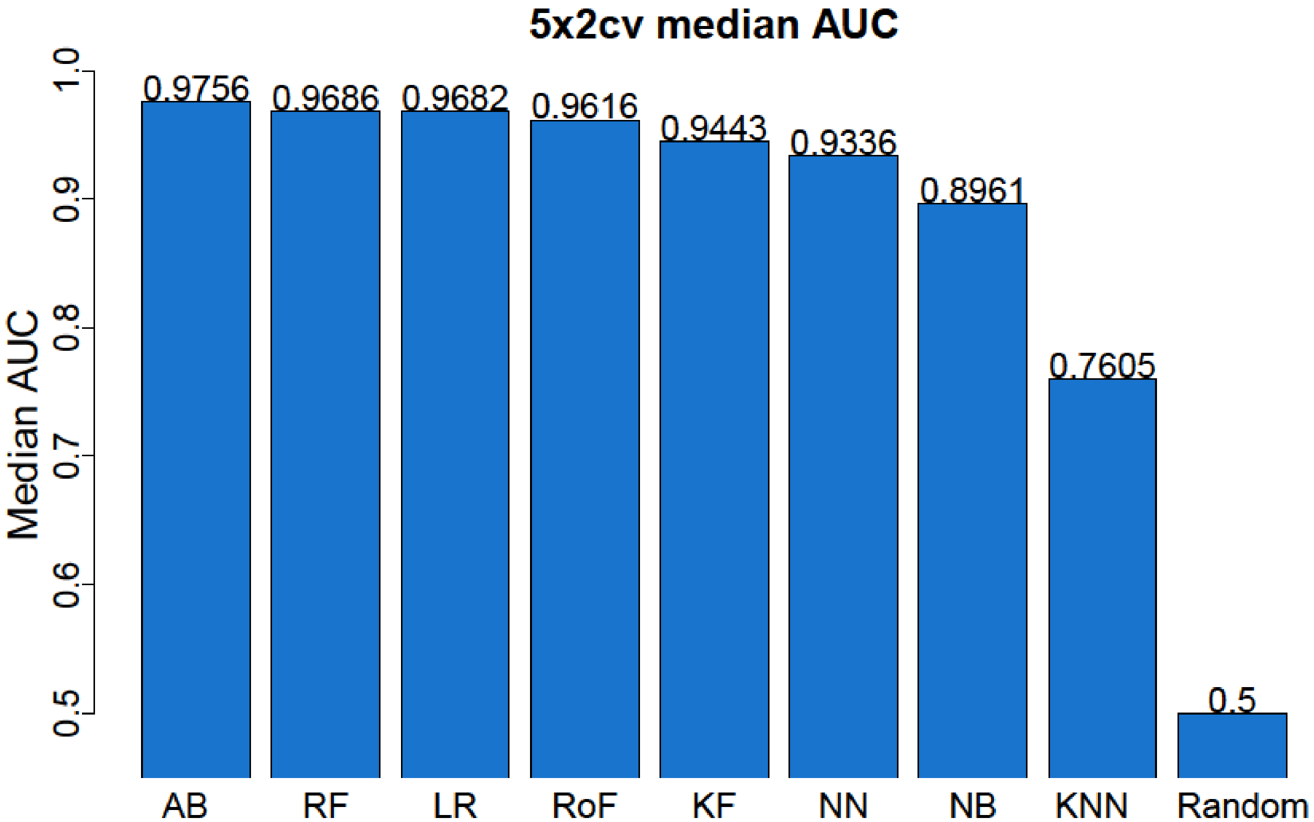
- Prediction algorithms
  - Single classifiers: k-nearest neighbors, naive Bayes, logistic regression, neural networks
  - Ensembles: random forest, adaboost, kernel factory, rotation forest

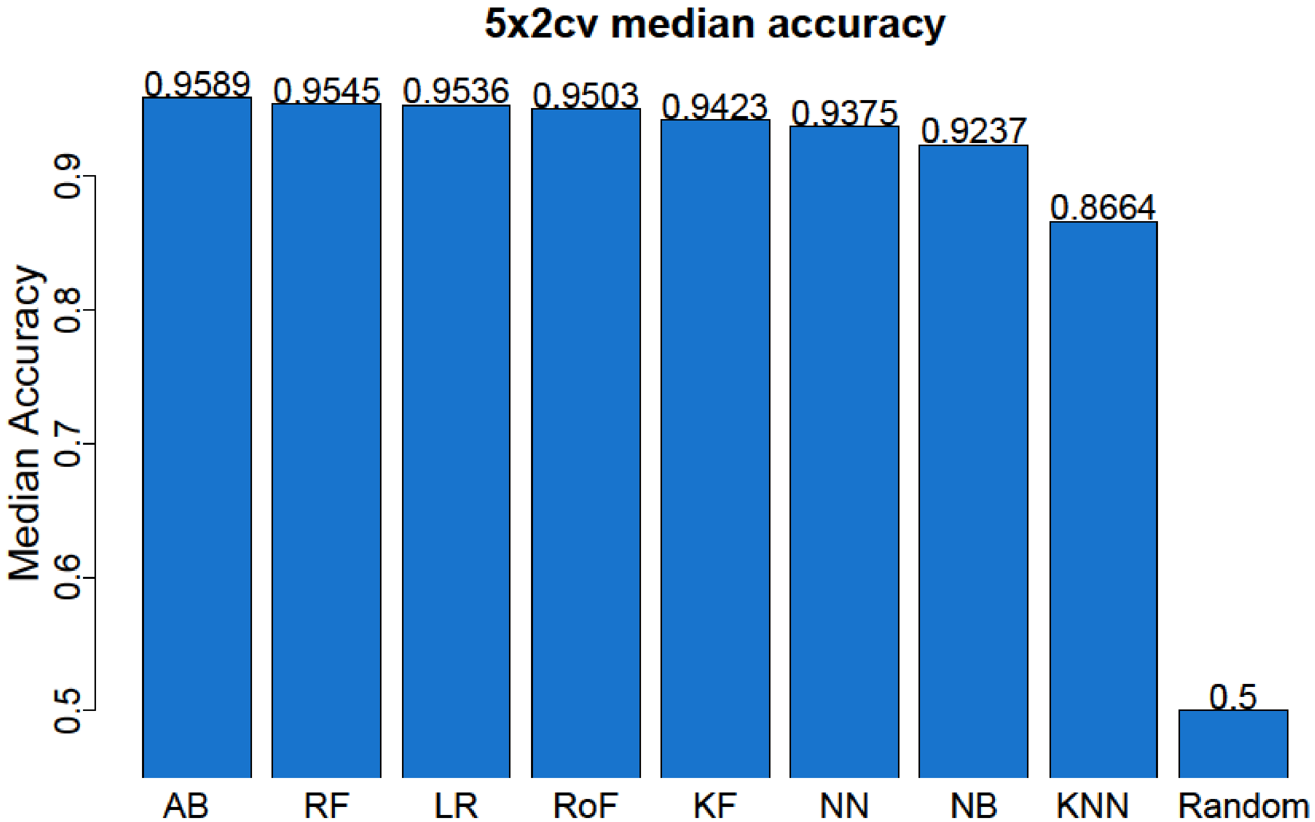
- Performance evaluation
  - G-mean, F1-measure, AUC, accuracy

- Cross-validation and tests
  - 5x2folds cross-validation
  - Friedman test with Bonferroni-Dunn post hoc tests

- Variable importance and partial dependence plots
  - Information-fusion sensitivity analysis

RESULTS: PREDICTION





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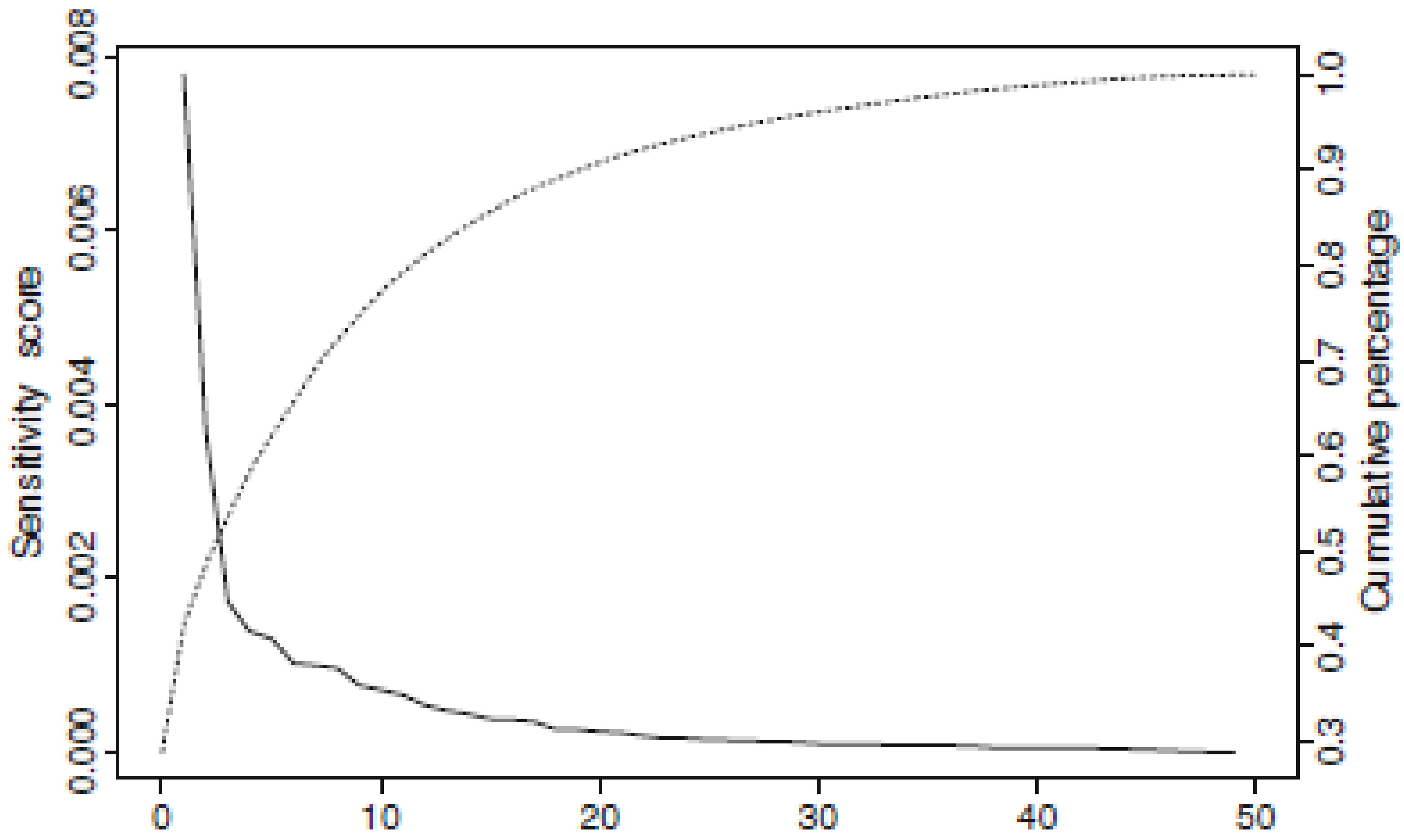
RESULTS: DISAGGREGATED FEATURES



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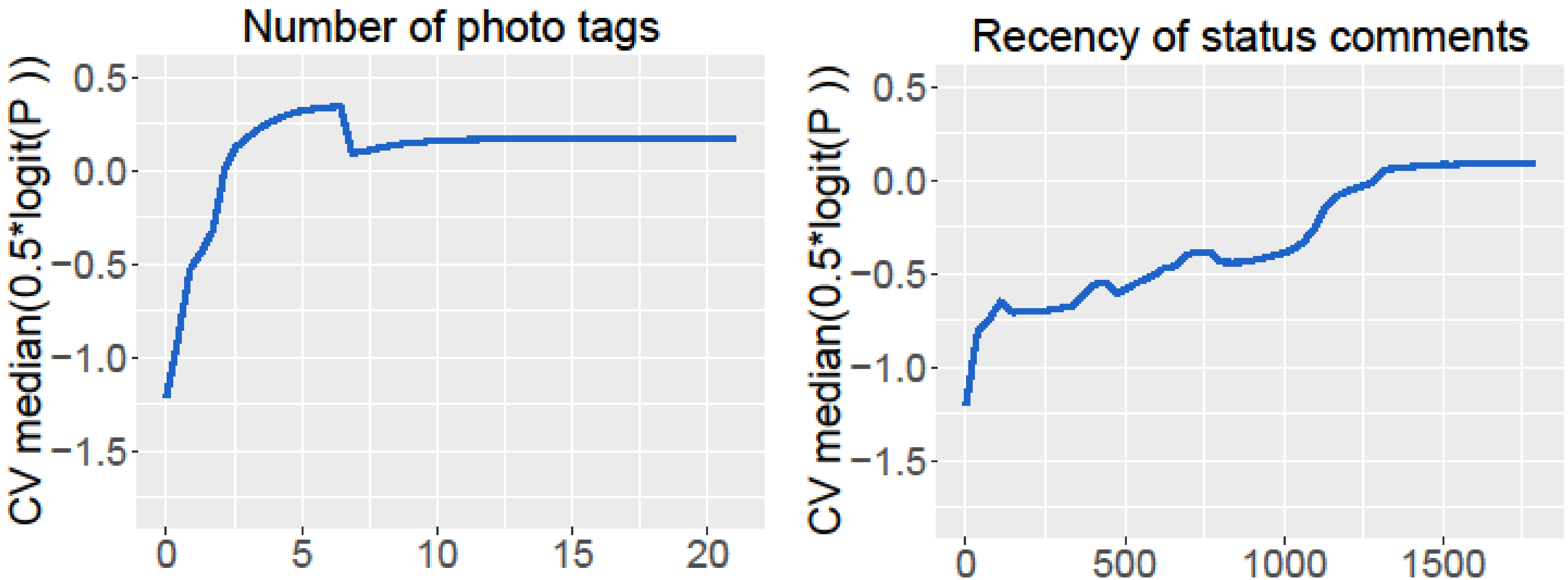
RESULTS: DISAGGREGATED FEATURES

Rank	Variable name	Median average sensitivity index	Type
1	IND (common gender)	0.0078	D
2	Age difference	0.0037	D
3	COUNT (tags on photos)	0.0017	F
4	IND (common location)	0.0014	D
5	COUNT (likes on statuses)	0.0013	F
6	REC (comments on statuses)	0.0010	T
7	REC (tags on photos)	0.0010	T
8	COUNT (common likes)	0.0010	P
9	COUNT (likes on photos)	0.0008	F
10	COUNT (common groups)	0.0007	P
11	REC (comments on photos)	0.0007	T
12	REC (tags on videos)	0.0005	T
13	REC (comments on albums)	0.0005	F
14	COUNT (tags on check-ins)	0.0004	F
15	COUNT (comments on photos)	0.0004	F
16	REC (comments on videos)	0.0004	T
17	COUNT (likes on photos)	0.0004	F
18	COUNT (comments on statuses)	0.0003	F

RESULTS: DISAGGREGATED FEATURES



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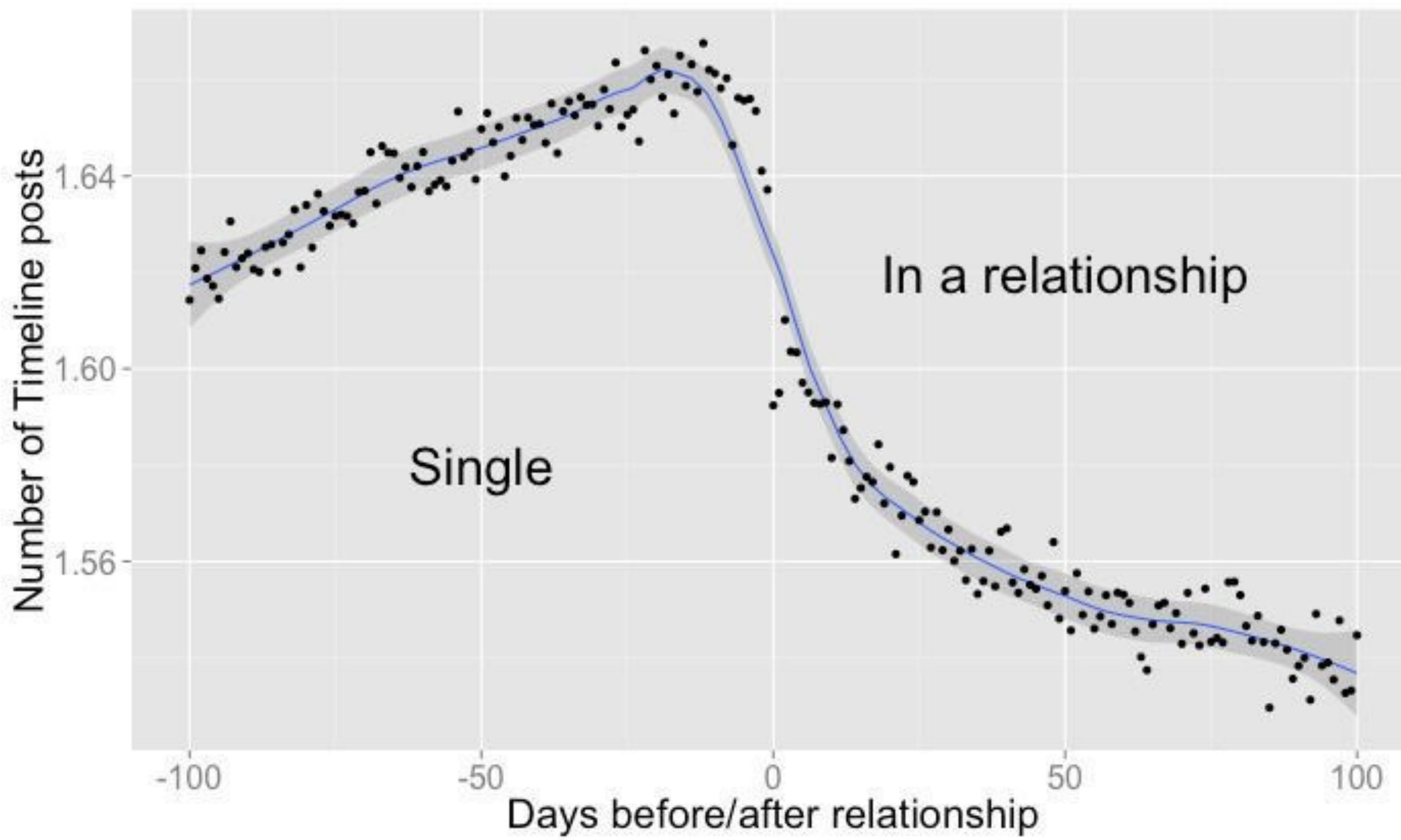
RESULTS: DISAGGREGATED FEATURES



Source: http://www.telegraph.co.uk/technology/facebook/10643858/Facebook-knows-when-youre-about-to-start-a-relationship.html

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DONATION BEHAVIOR

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INTRODUCTION

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Predicting donation behavior: Acquisition modeling in the nonprofit sector using Facebook data

Lisa Schetgen, Matthias Bogaert *, Dirk Van den Poel

Ghent University, Department of Marketing, Tweekerkenstraat 2, 9000 Ghent, Belgium

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ABSTRACT

The purpose of this study is to demonstrate the value of Facebook data in predicting first-time donation behavior. More specifically, we provide evidence that Facebook data can be used as a valuable data source for nonprofit organizations in acquiring new donors. To do so, we evaluate three different dimensionality reduction techniques (i.e., singular value decomposition, non-negative matrix factorization, and latent Dirichlet allocation) over seven classification techniques (i.e., logistic regression, k-nearest neighbors, bagged trees, random forest, adaboost, extreme gradient boosting, and artificial neural networks) using five times twofold cross-validation. Next, we assess what type of Facebook data and which predictors are most important. The results indicate that we can predict first-time donation behavior based on Facebook data with high predictive performance. Our benchmark indicates that the combination of singular value decomposition and logistic regression outperforms all other analytical methodologies with an area under the receiver operating characteristic of 0.72 and a top decile lift of 3.33. The results show that Facebook pages and categories of Facebook pages are the most important data types. The most important predictors are dimensions related to age, education, residence, materialism, responsible consumption, and interest in nonprofits. The presented acquisition models can be used by nonprofit organizations to implement a one-to-one targeted marketing campaign towards Facebook fans. To the best of our knowledge, our study is the first to determine the predictive value of Facebook data for nonprofits in a real-life acquisition context.

INTRODUCTION

- Predicting who is most likely to become a donor. Why?
  - Targeting the right people
  - Limited customer lifetime
  - Increased fundraising results

- Predicting based on social media data. Why?
  - Nonprofits remain unsure about how to incorporate social media into CRM
  - To infer personal characteristics
  - Value of social media is proven in other prospecting contexts

INTRODUCTION

TRADITIONAL DONOR ACQUISITION → INCORPORATION OF SOCIAL MEDIA

TRADITIONAL DONOR ACQUISITION

- Inherently difficult because there is no pre-existing relationship
- External databases or market research agencies

INCORPORATION OF SOCIAL MEDIA

- Facebook fans publicly display their interest
- Identification of a substantial base of potential donors (i.e., Facebook fans)
- Does not require a pre-existing relationship

RESEARCH QUESTIONS

1. Is it *possible* to predict donation behavior using solely social media data?

2. What *type of social media data* matters the most?

3. Which *variables* are most important?

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LITERATURE OVERVIEW

Table 1
Overview of CRM literature concerning the nonprofit sector, as well as studies regarding donation behavior using social media.

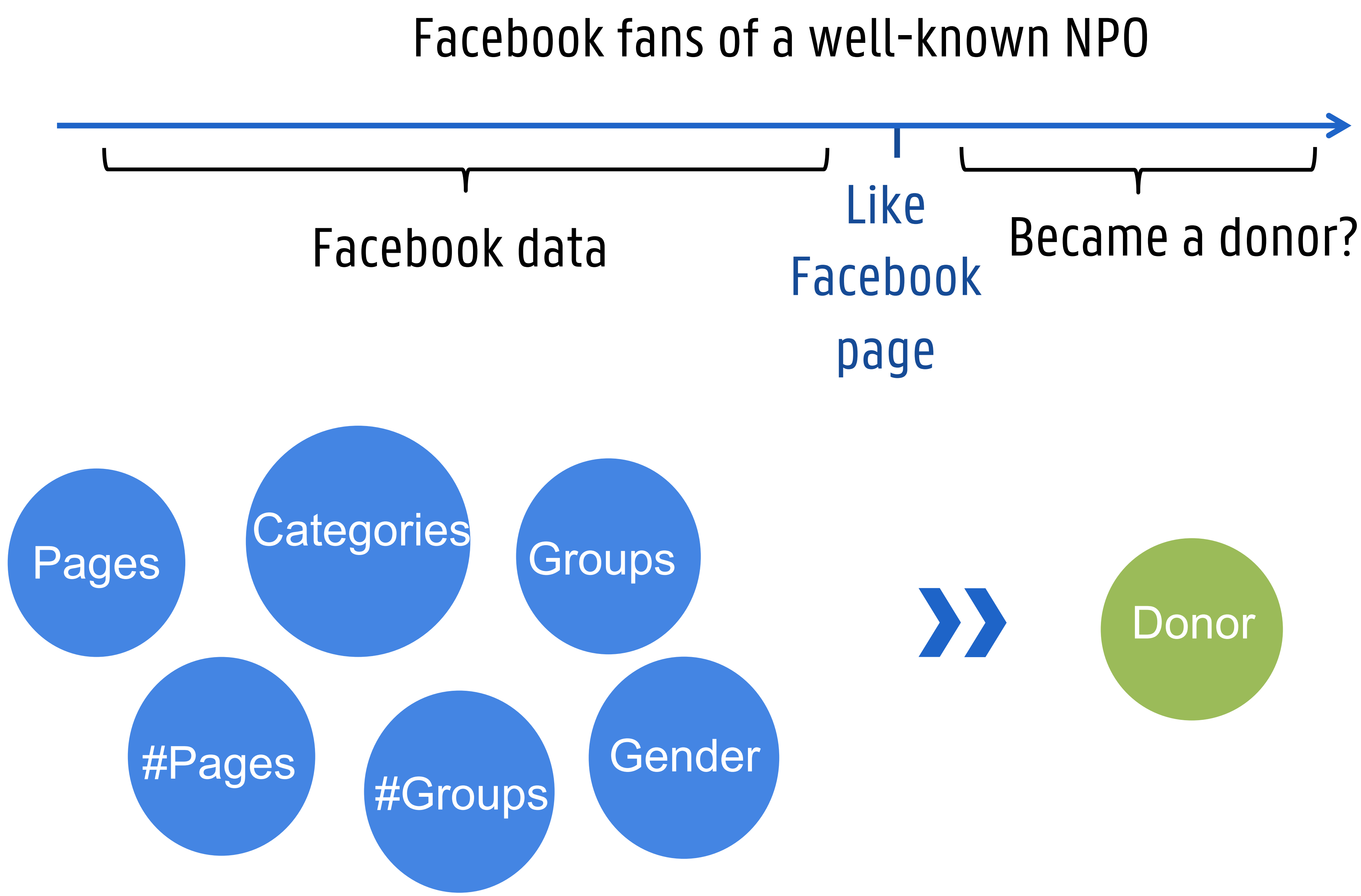
Study	Research type		CRM domain		Data type	
	Diagnostic	Predictive	Retention	Acquisition	Traditional	Social media
Sargeant [52]	x		x		x	
Bennett [7]	x		x	x	x	
Hsu et al. [35]	x		x	x	x	
Germain et al. [32]	x		x		x	
Ferguson [27]	x		x	x	x	
Schlumpf et al. [53]	x		x		x	
Garner & Garner [31]	x		x		x	
Enjolras et al. [24]	x				x	x
Chell & Mortimer [15]	x		x		x	x
Brown & Taylor [12]	x				x	x
Courtois & Verdegem [17]	x				x	x
Warren et al. [60]	x					x
Farrow & Yuan [26]	x				x	x
Godin et al. [33]		x	x		x	
Althoff & Leskovec [2]		x	x		x	
Lee & Chang [41]		x	x	x	x	
Verhaert & Van den Poel [59]		x	x	x	x	
Our study		x		x		x

LITERATURE OVERVIEW

- Literature gap:
  - No predictive model on the basis of solely social media data for acquisition in the nonprofit sector

- Contributions:
  - Predictive study in a real-life acquisition context
  - Crawled social media data for predictive rather than diagnostic research

DATA



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DATA

- Identify list of **potential donors**
  - Scrape list of Facebook fans using RSelenium

- Compute **dependent variable**
  - Use internal database to determine which Facebook fans became a donor *after* liking the Facebook page
  - Match based on names
  - Only 1.23% are donors (96 out of 7795)
    - Inherent to the issue at hand
    - Random oversampling to deal with class imbalance

DATA

- Collect and compute **independent variables**
  - For every Facebook fan, the following data is scraped using RSelenium:
    - Liked Facebook pages
    - Joined Facebook groups
    - Gender
  - From this data, we compute:
    - User-page matrix
    - User-category matrix
    - User-group matrix
    - Gender (missing values are imputed using Genderize.io API)
    - Total number of liked pages
    - Total number of joined groups

DATA

- User-page, user-category and user-group matrices:
  - Element x_{ij} of the user-page matrix is set to 1 if user i likes Facebook page j.
  - Issue: 68% (78%) of the pages (groups) are liked by only 1 individual
    - Only keep pages, categories and groups that have been liked by at least 20 users in our sample
    - Nevertheless: Dimensionalities and sparsity levels remain high
    - Use dimensionality reduction techniques on each of the matrices separately to distinguish between the different types of Facebook features

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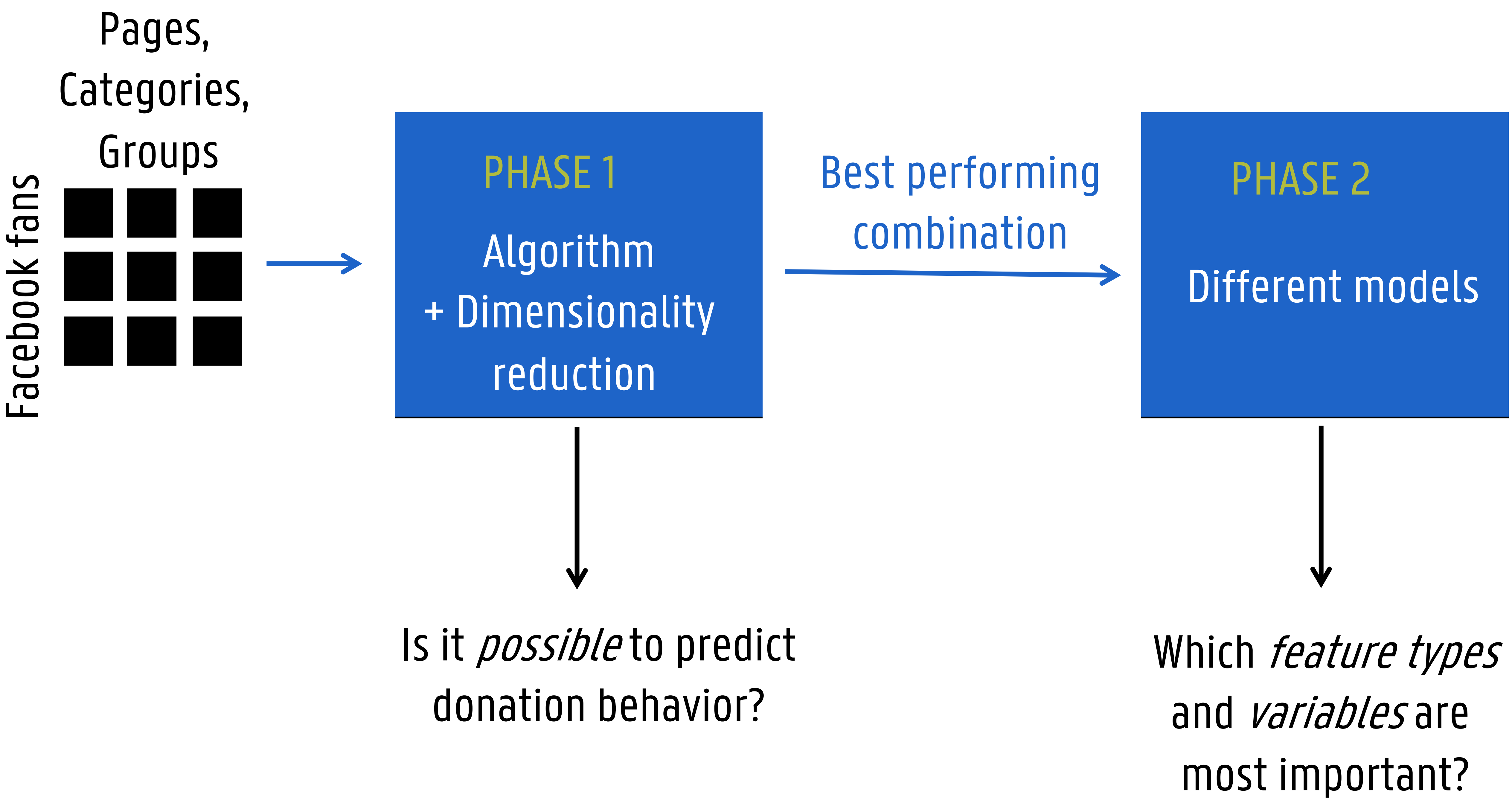
METHODOLOGY



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METHODOLOGY: PHASE 1

- Dimensionality reduction (DR) techniques:
  - None = Baseline model
  - Singular Value Decomposition (SVD)
  - Non-negative Matrix Factorization (NMF)
  - Latent Dirichlet Allocation (LDA)
- Prediction algorithms:
  - Single classifiers: logistic regression, k-nearest neighbors, neural network
  - Ensembles: bagged trees, random forest, adaboost, xgboost
- Performance evaluation:
  - AUC
  - Top Decile Lift (TDL)
- 5x2cv for every combination of algorithm, DR technique and number of dimensions k (50, 100, and 200)

METHODOLOGY: PHASE 2

- To assess the value of the distinct types of Facebook data:
  - Models on different subsets of features

	Description	M1	M2	M3	M4	M5	M6	M7	M8
Pages									
	100 dimensions related to Facebook pages, resulting from the application of a dimensionality reduction technique on the user-page matrix	x			x	x	x		x
Categories									
	100 dimensions related to categories of Facebook pages, resulting from the application of a dimensionality reduction technique on the user-category matrix		x		x	x		x	x
Groups									
	100 dimensions related to Facebook groups, resulting from the application of a dimensionality reduction technique on the user-group matrix			x	x		x	x	x
Frequency									
	#Liked pages (total number of Facebook pages) and #Groups (total number of Facebook groups)					x	x	x	x

Individual value   Added value

- To determine the most important variables:
  - Absolue value of the β coefficients (LASSO regression)
  - Underlying pages, categories and groups of top 10 most important variables (i.e., dimensions)

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RESULTS: PHASE 1

	AUC				TDL			
	Binary	SVD	NMF	LDA	Binary	SVD	NMF	LDA
LR	0.6658	**0.7239 *(k=100)***	0.6960 *(k=100)*	0.6989 *(k=100)*	2.6265	**3.3305**	2.9500	2.7250
KNN	0.6326	0.6078 *(k=100)*	**0.6408 *(k=100)***	0.6263 *(k=50)*	**2.0770**	1.5125	1.9980	1.7215
BT	0.5898	0.6155 *(k=100)*	0.6114 *(k=50)*	**0.6532 *(k=100)***	1.7130	1.6020	1.5550	**1.7700**
RF	0.6663	**0.6820 *(k=100)***	0.6735 *(k=50)*	0.6757 *(k=100)*	**2.4185**	1.8185	2.0020	2.1515
AB	0.6387	**0.6639 *(k=100)***	0.5496 *(k=50)*	0.5985 *(k=50)*	**2.3645**	1.8595	1.1920	1.5935
XGB	0.5978	**0.6630 *(k=100)***	0.5833 *(k=50)*	0.6463 *(k=50)*	1.7685	1.8300	1.2970	**2.0145**
NN	0.5854	0.6793 *(k=200)*	0.6624 *(k=200)*	**0.7009 *(k=50)***	**2.2985**	2.1365	1.9790	2.2080

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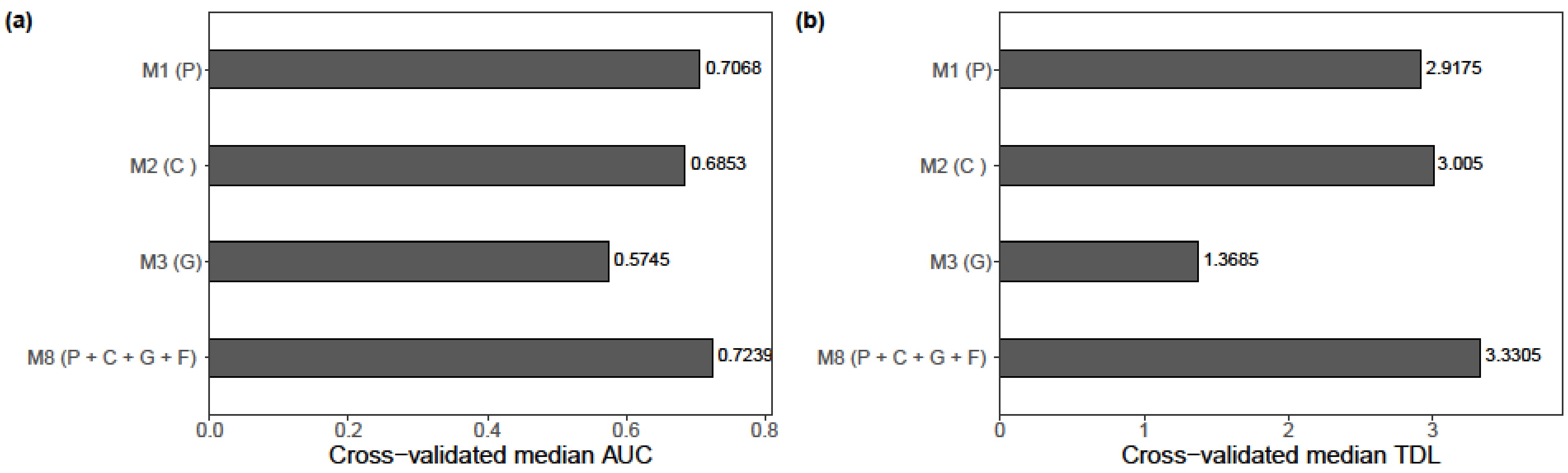
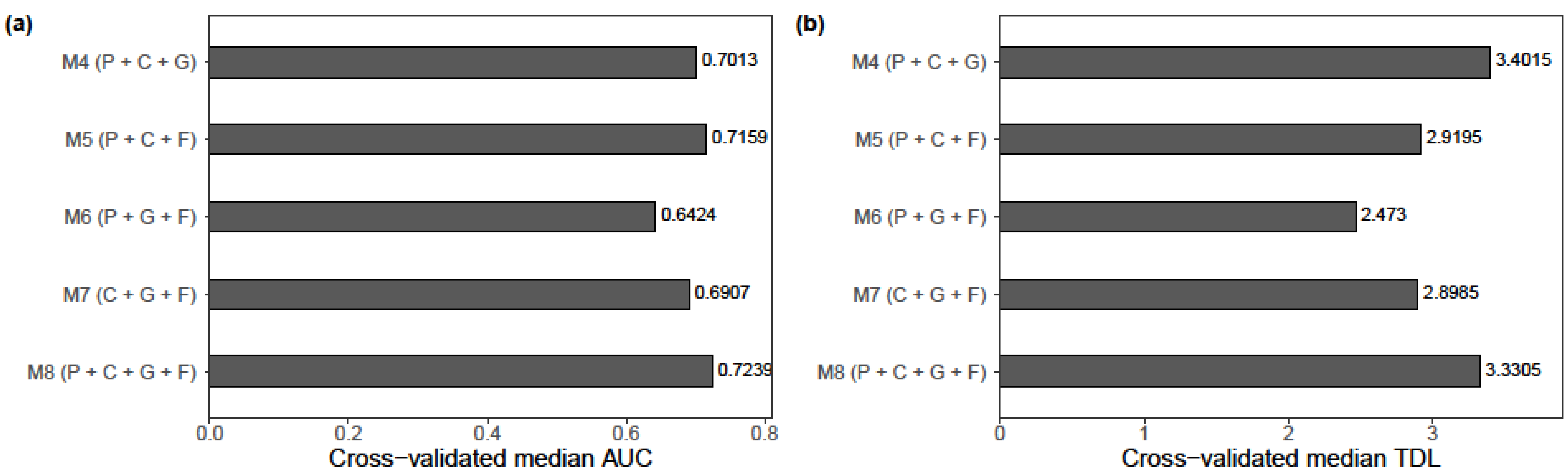
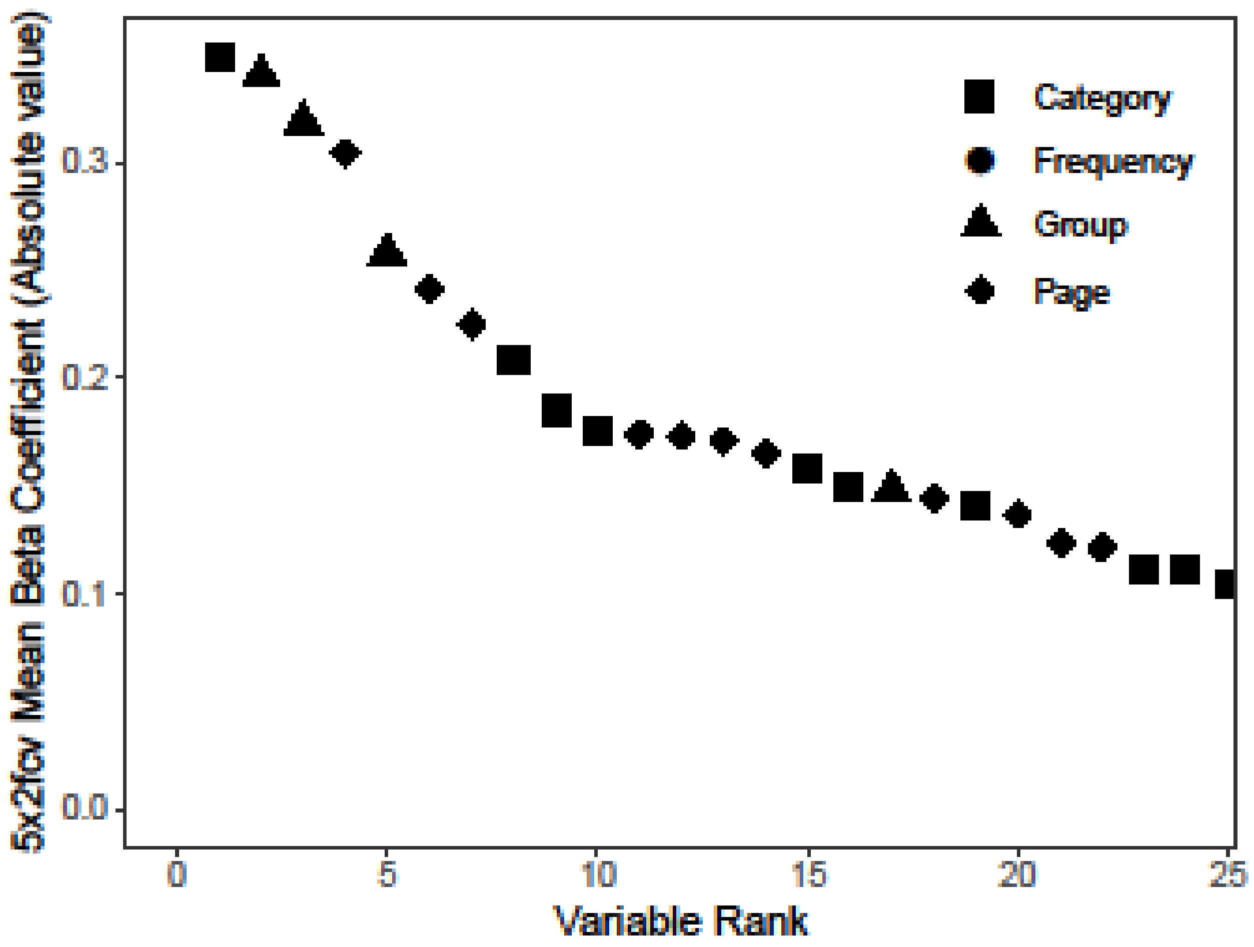
RESULTS: PHASE 2





RESULTS: PHASE 2



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RESULTS: PHASE 2

1. Category	2. Group	3. Group	4. Page	5. Group
Bar	Namur, Belgium	Police Control Belgium (FR)	Studio Brussel *(radio station)*	VIRTUAL FLEA MARKET LIEGE (FR)
Dance- and nightclub	2nd hand clothes NIVELLES (FR)	Info radars, police controls Liège (FR)	Stromae	Zoo of Antwerp (NL)
Pub	Selling, buying services in Brussels (FR)	You are a real Liégois if…(FR)	Decovry.com - Be the first to discover	For sale Brussels (FR)
Publisher	2nd hand and new items for sale, Antwerp (NL)	Recommended restaurants in Liège (FR)	De Ideale Wereld *(tv show)*	Looking for apartment(-sharing) Brussels (FR)
Public service	Bart's way (NL)	Liégeoises Liégeois (FR)	Doctors Without Border (Belgium) (NL)	JOBS BELGIUM (SPA)
Café	Halle Selling, Free or Trading group (NL)	Radars Liège (FR)	Humo *(magazine)*	Info radars and controls-Charleroi (FR)
Education	For sale in the region of Mons (FR)	Most beautiful pictures of Liège (FR)	Nutella	Green bio zero waste (FR)
Higher education and university	For sale in Mons and its environs (FR)	Bob from Liège (FR)	De Morgen *(newspaper)*	Bouddhism
Business service	All cars for sale in Belgium (FR)	Gastronomy from Liège (FR)	UNICEF Belgium	Second hand books Brussels (FR)
Jewelry/watches	AJITEBADLO.COM	Police, Accidents, Road Conditions (FR)	UNILAD	2ND HAND CLOTHES CHARLEROI (FR)

6. Page	7. Page	8. Category	9. Category	10. Category
Sneaker District	Doctors Without Borders (Belgium) (FR)	Touring company	Organization	Event planner
Decovry.com - Be the first to dis	Amnesty International Belgium (FR)	Business service	Public organization	Hairdresser
Guts & Gusto	UNICEF Belgium	Café	Society- and culturewebsite	Society- and culturewebsite
CLUSE	Amnesty International	Magazine	Cosmetics store	Education
Loavies	GuiHome vous détend *(humorist)*	Hairdresser	Health- and wellnesswebsite	Youth organization
Colourful Rebel	Doctors Without Borders (FR)	Publisher	Magazine	Library
Coca-Cola	UNICEF	Performance Art	Public service	Non-governmental organization (NGO)
TravelBird	Barack Obama	Higher education and university	Other	Dance- and nightclub
LADbible	Doctors Without Border (Belgium) (NL)	Medical company	Movie-/tv-studio	Sports
EF Belgium - Study abroad	Doctors of the World Belgium (FR)	Telecommunication company	Entrepreneur	Fastfoodrestaurant

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CONCLUSION

- It *is possible* to predict which Facebook fans are most likely to become donors
  - Decision support system during CRM efforts: who to target

- Facebook *pages and categories* are most important

- *Insights* can be gained by inspecting the most important dimensions

→ Demonstration of the value and potential of social media data to nonprofits

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FACULTY OF ECONOMICS AND
BUSINESS ADMINISTRATION

SOCIAL MEDIA AND WEB ANALYTICS

Prof Dr Matthias Bogaert

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